

Chinese-SkillSpan: a benchmark for competency span extraction in Chinese job advertisements

Guojing Li^{†,2}, Zichuan Fu^{†2}, Wenlin Zhang², Kaifeng Guo², Jinning Yang², Xinyang Wu³, Jingtong Gao², Junyi Li², and Xiangyu Zhao²

¹Renmin University of China, Beijing, China

²City University of Hong Kong, Kowloon, Hong Kong SAR, China

³Wuhan University, Wuhan, Hubei, China

Corresponding author:

Xiangyu Zhao

Email address: xianzhao@cityu.edu.hk

ABSTRACT

Background. Chinese job advertisements express competencies through coordinated phrases and context-dependent terms. Comparing extraction systems requires consistent decisions about which characters form each mention and which competency type it expresses.

Methods. We developed the Chinese-SkillSpan benchmark from a corpus of 22,840 sentences across four recruitment collections. Character-level annotations distinguish language, knowledge, occupational skills, and transversal competences. An ESCO-informed handbook was refined through human coding, disagreement analysis, and adjudication to specify annotation scope, boundaries, and types. The resource separates a frozen human reference from LLM-assisted Silver supervision. Silver checks combined automated structural validation with human review of samples before and after bulk annotation. Three annotators independently applied the final handbook to a separate random sample, blinded to machine suggestions and one another’s labels. We compared prompted models, supervised adaptation, and multilingual encoders.

Results. Silver-supervised Qwen adaptation improved typed exact micro-F1 on the human reference from a no-adaptor baseline of 0.361 to 0.540 ± 0.035 (mean $\pm$ sample SD across three training seeds) under fixed inference rules. Each adapted model also improved exact recall across all assessed span-length groups. Independent blinded coding showed higher agreement on boundaries alone than on typed spans, indicating additional difficulty in type assignment.

Conclusions. Chinese-SkillSpan pairs source-text spans with explicit annotation rules and separate human and Silver labels, enabling model comparisons and inspection of boundary and type errors. Human-reference results concern data used in guideline development; generalization to unseen advertisements requires independent testing.

INTRODUCTION

Online job advertisements describe the knowledge and skills employers seek and provide evidence of changing occupational requirements (Podjanin and Strietska-Ilina, 2020; Senger et al.,

[†]Guojing Li and Zichuan Fu contributed equally to this work.

2024). Document-level methods retrieve skills from predefined inventories (Bhola et al., 2020), whereas span extraction locates the words expressing a requirement (Gnehm et al., 2022a). Job Skill Named Entity Recognition (JobSkillNER) identifies these mentions and assigns a type to each span. Preserving source wording makes boundary and type assignments inspectable. Even short Chinese phrases can require several annotation decisions. A technology name can denote tool use in one clause and knowledge of principles in another. Coordination may express separate activities or share an action that must remain attached to its objects. Chinese normally lacks word spaces, so tokenization alone cannot resolve these semantic boundaries (Peng and Dredze, 2015; Chen and Kong, 2021). Similar ambiguities occur in other languages (Nguyen et al., 2024), but dataset translation does not determine which Chinese substrings to select. Explicit context-sensitive rules are needed; Figure 4 pairs Chinese examples with English explanations. Existing resources provide precedents, but their annotation schemes differ. English SkillSpan permits overlapping SKILL and KNOWLEDGE layers (Zhang et al., 2022b). Kompetenzer adds ESCO-derived fine-grained classification to Danish and English spans (Zhang et al., 2022a). German recruitment work extracts education, experience, and language requirements before finer classification (Gnehm et al., 2022a). Document-level retrieval and Chinese skill-demand forecasting use other prediction targets (Bhola et al., 2020; Chen et al., 2024). We introduce Chinese-SkillSpan, a benchmark for competency span extraction in Chinese job advertisements. Built from a corpus of 22,840 sentences across four recruitment collections, it combines annotation guidelines, distinct supervision and reference subsets, scoring protocols, and model baselines. The handbook distinguishes language (L), knowledge (K), occupational skills (S), and transversal competences (T). In Chinese-SkillSpan, we use *competency spans* for contiguous source-text mentions across all four categories and *occupational skills* specifically for S. The scheme follows ESCO’s high-level categories and the knowledge–skill distinction in the European Qualifications Framework (European Commission, 2024; Council of the European Union, 2017), without assigning ESCO concept identifiers. Character offsets preserve each mention’s wording. The study addresses three linked questions: how to delimit and type Chinese competency mentions, how consistently annotators apply those rules, and how extraction models perform under the resulting conventions. The handbook supplies boundary and context rules with positive and negative examples. A separate three-annotator blinded study measures agreement under the final handbook, while historical coding and assisted review document reference construction and Silver quality checks. Matched Qwen adaptation and Chinese-supervised encoder comparisons assess model performance, with diagnostics for type, span length, and empty-reference sentences. The original corpus and its smaller training, reference, and agreement subsets have distinct roles. The 150-sentence human reference contributed to guideline development, so model comparisons on it are diagnostic. External recruitment experiments provide additional task-specific comparisons. We document each subset’s role and availability following guidance on datasets and annotation agreement (Bender and Friedman, 2018; Gebru et al., 2021; Artstein and Poesio, 2008).

RELATED WORK

Skill extraction and recruitment datasets. Recruitment datasets differ in prediction targets. Sayfullina et al. (2018) ask whether a candidate soft-skill phrase refers to an applicant, whereas Bhola et al. (2020) retrieve skills from a predefined inventory for an entire advertisement. Span extraction locates requirement wording. Within that task, SkillSpan annotates English SKILL

and KNOWLEDGE in two potentially overlapping layers; hard and soft skills describe its coverage rather than its label inventory (Zhang et al., 2022b). Kompetencer combines manual skill and knowledge spans with ESCO-derived fine-grained labels in Danish and English; its distantly supervised Danish test labels were manually corrected (Zhang et al., 2022a). Gnehm et al. (2022a) extract coarse EDU/EXP/LNG spans before refining skill areas for taxonomy matching. Chinese recruitment research includes both forecasting and entity extraction. Job-SDF evaluates skill-demand forecasting at several aggregation levels (Chen et al., 2024). For occupational profiling, Cao et al. (2021) manually annotated skill and specialty requirements using BIOES and trained a BERT–BiLSTM–CRF model. Chinese-SkillSpan instead uses flat character spans with four types and explicit rules for coordination and contextual interpretation. Taxonomy-based weak supervision can reduce manual annotation work (Zhang et al., 2022c), but generated labels still need validation. Nguyen et al. (2024) report difficulties with conjoined and ambiguous mentions across six recruitment datasets. Senger et al. (2024) also identify scarce public annotations and inconsistent terminology. Table 1 compares the prediction units, annotation layers, and available materials of selected resources.

Table 1. Selected recruitment resources: prediction units, annotation layers, and reported materials. Table 2 details Chinese-SkillSpan counts and data roles.

Work / language	Prediction unit	Actual annotation layers	Material scope
Sayfullina et al. (2018); English	Candidate phrase in context	Relevance to an applicant; binary disambiguation, not open-boundary extraction	Annotated soft-skill contexts
Gnehm et al. (2022a); German	Spans, then finer areas	Coarse EDU/EXP/LNG; finer skill areas and taxonomy matching	Article and model; full annotated corpus access not established here
Zhang et al. (2022b); English	Source-text spans	SKILL / KNOWLEDGE; overlap permitted between the two layers	Dataset, code, and annotation guidance
Zhang et al. (2022a); Danish / English	Spans and fine-grained labels	Manual skill/knowledge spans; ESCO API distant labels; Danish test-label correction	Dataset and code
Cao et al. (2021); Chinese	Recruitment entities	Manually annotated skill / specialty entities; BIOES	Article and supporting data; full NER corpus access not established here
Chinese-SkillSpan; Chinese	Flat character spans	L/K/S/T; frozen human reference and Silver supervision; separate blinded agreement study	Handbook, protocols, baseline predictions, scorer and independent coding exports (Appendix D)

Chinese span annotation and competency guidelines. Chinese NER resources cover social-media text (Peng and Dredze, 2015) and fine-grained entity categories (Xu et al., 2020). Multilingual studies examine recognition in noisy text (Fetahu et al., 2023) and changes in contextual entity types (Luo et al., 2025). These tasks provide background but do not specify how to divide

recruitment requirements into LSKT spans. Chinese boundary detection is complicated by the absence of explicit word delimiters (Chen and Kong, 2021); recruitment annotation also distinguishes knowledge, occupational actions, and transversal competences in each clause. ESCO provides a multilingual vocabulary for labor-market information (le Vrang et al., 2014; European Commission, 2024). SkiLLens combines human involvement, skill extraction, and ESCO linking across European job advertisements (De Santo et al., 2026). Chinese-SkillSpan adapts ESCO’s high-level categories to LSKT annotation in Chinese. The handbook specifies when tool use denotes S rather than K, when coordinated objects must retain a shared action, and when an experience qualifier belongs inside the span. Agreement analysis and adjudication consequently address both unitization and categorization (Artstein and Poesio, 2008).

Model-assisted annotation and extraction models. LLMs can supply first-pass labels, with their usefulness depending on task instructions and human validation (Ding et al., 2023; Li et al., 2023; Ul Haq et al., 2026). GoLLIE and subsequent guideline-refinement work study how annotation instructions shape model outputs (Sainz et al., 2024; Kim et al., 2026). In recruitment, JobScape generates synthetic advertisements for skill-to-taxonomy matching (Magron et al., 2024). Chinese-SkillSpan retains collected recruitment texts and adds model-generated annotations with documented review. The annotation prompts follow the handbook used at each stage. Encoder models offer one way to learn from these labels. Pretrained encoders and structured decoding (Devlin et al., 2019; Lafferty et al., 2001) can be combined with continued pretraining on recruitment text (Gururangan et al., 2020; Zhang et al., 2022b). SkillSpan introduced the domain-adapted JobBERT and JobSpanBERT baselines (Zhang et al., 2022b). ESCOXML-R uses taxonomy-driven multilingual pretraining (Zhang et al., 2023), and NNOSE retrieves training-data representations to support extraction, including infrequent patterns (Zhang et al., 2024). Reproducing these methods on their original tasks and adapting them to Chinese LSKT extraction require separate protocols because the languages, annotation layers, and splits differ. Other approaches use text-to-structure generation in UIE (Lu et al., 2022), distilled supervision in UniversalNER (Zhou et al., 2024), open-type encoder extraction in GLiNER (Zaratiana et al., 2024), and generative extraction with self-verification in GPT-NER (Wang et al., 2025). These approaches provide methodological context; they are not additional evaluated baselines. The models tested here, their supervision, and their scoring protocols are specified in Evaluation Protocol.

CORPUS AND ANNOTATION

Task Definition and Label Schema

Given a Chinese job-ad sentence x , the task is to predict non-overlapping spans (s, e, t) . Here, s and e are zero-based, end-exclusive Unicode-code-point offsets, and $t \in \{L, K, S, T\}$. Each span must reproduce the contiguous substring $x[s : e]$, including its spelling and internal spaces. The annotation describes a requirement in the advertisement; it does not measure an applicant’s ability. The handbook assigns L to language requirements, including language names, proficiency levels, examinations, and certificates. K covers facts, principles, theories, and fields of study. S covers occupational actions, methods, and tool use. T covers transversal capabilities such as communication, cooperation, cognition, and self-management. These categories draw on ESCO and on the distinction between knowledge and skills in the European Qualifications Framework (European Commission, 2024; Council of the European Union, 2017).

Figure 1 shows how corpus preparation and annotation lead to model training and evaluation.

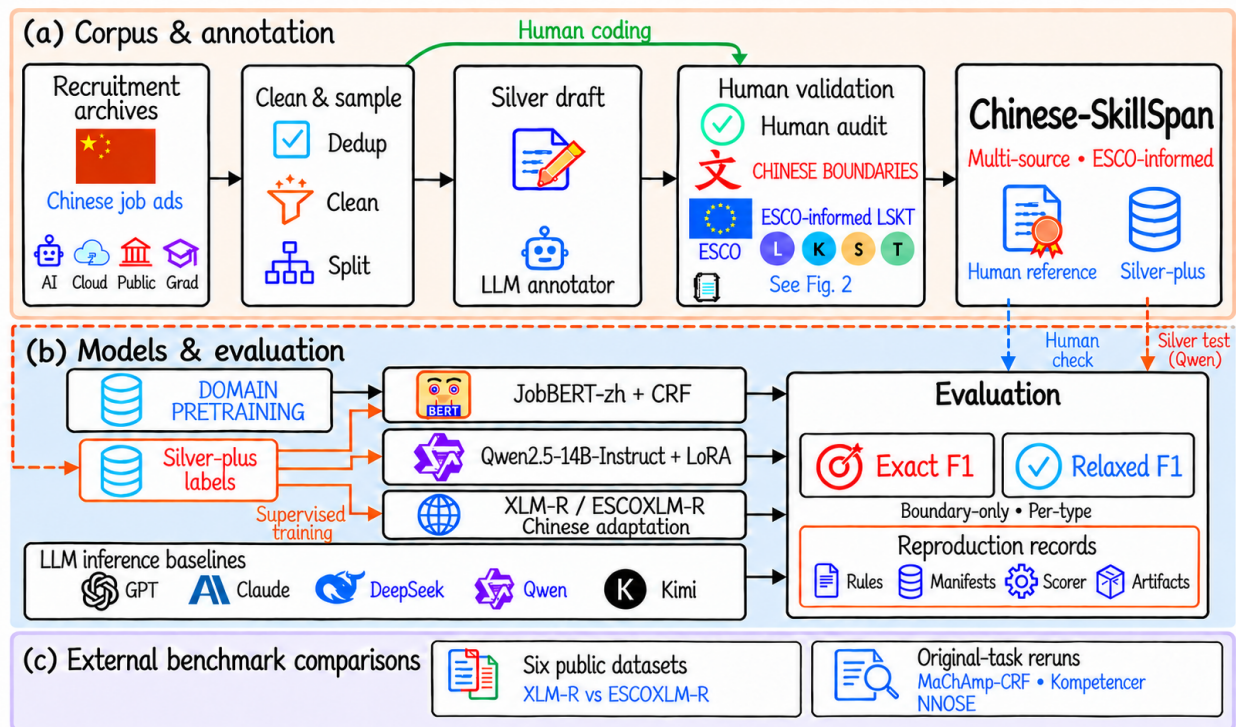


Figure 1. Chinese-SkillSpan workflow: corpus preparation, annotation, models, and evaluation. Human-reference checks and Silver-target evaluation are distinguished. External comparisons retain separate datasets, labels, and metrics. Figure 2 separates reference construction, assisted review, and the independent blinded agreement study.

By project convention, educational and non-language qualifications also receive K. They serve as recruitment proxies for knowledge requirements, so K counts combine substantive knowledge mentions with qualification proxies. We have not established separate proxy counts or evaluated the task after excluding them. The rules below explain how context determines type and how experience requirements are handled. Evaluation concerns mention boundaries and LSKT types. Figure 2 separates reference construction, Silver annotation and review, and the independent blinded agreement study. Table 2 lists their sample sizes and intended uses.

Corpus Construction and Annotation Layers

The corpus contains 22,840 sentences from the AI, Cloud, public-sector, and graduate-recruitment collections, with an original split of 17,460 training, 2,143 development, and 3,237 test sentences. These collection names describe sources and can overlap in occupational coverage. Table 2 distinguishes the corpus from the smaller subsets used for annotation, supervision, and evaluation. Texts came from three routes: commercial collections purchased from MacroData (MacroData, nd); public-institution recruitment announcements collected by the research team; and the Recruitment Dataset (dataset ID 163746) on Alibaba Cloud Tianchi (Tianchi, nd). For public-institution announcements, team members selected pages from government portals, university and hospital websites, and public recruitment platforms across regions of China. A crawler revisited the reg-

Human annotation and LLM-assisted labeling

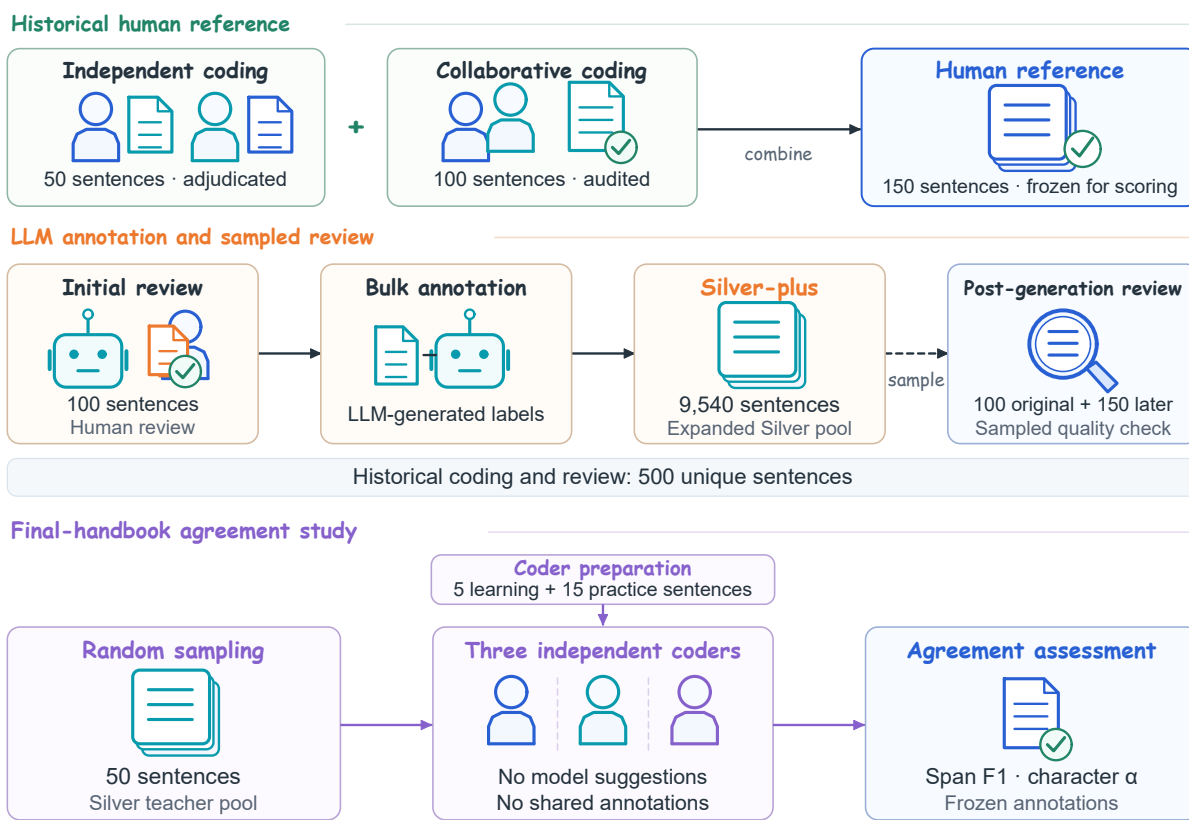


Figure 2. Distinct paths for reference construction, assisted labeling, and independent blinded agreement. The separate 50-sentence study follows training and uses three independent coders with machine and peer labels hidden. Agreement uses pre-discussion annotations. Historical coverage remains 500 unique sentences. The new sample assesses agreement under the final handbook; it neither extends the 150-sentence reference nor serves as a model test set.

istered URLs weekly or monthly. Screening and cleaning produced the public-institution subset. Appendix D records the available source metadata and gaps in the original collection history. The original annotation study contains 2,601 records: a 150-sentence human reference, named Gold150 in the released files, and 2,451 records with LLM-generated Silver labels, named Silver-plus. The Silver set includes 100 initially reviewed records, 730 further records from the model-disagreement queue, and 1,621 historical extraction records. The 980-record disagreement queue contributed the reference sentences and the first 830 Silver records. The human reference combines 50 calibration sentences with a non-overlapping 100-sentence challenge sample. The calibration sentences were coded independently from raw text and then adjudicated; the challenge sample was collaboratively coded and reviewed. Both informed guideline development. Their final annotations were fixed for evaluation. The reference contains 663 spans: 448 S, 125 K, 88 T, and two L. Eight sentences contain no annotated span. Its 150 complete sentence texts are distinct after NFC normalization. The two original 100-sentence review samples are separate from the reference, giving 350 unique sentences

with human coding or review at this stage. The historical 15-sentence assisted-review subset is nested within this coverage. Appendices B and D give cohort counts and sampling details.

Table 2. Data layers, support, and intended use. The reference and original Silver-plus pool partition the 2,601-record lineage; the expanded pool is a later experiment-specific construction. The two 100-sentence reviews are distinct subsets of Silver-plus; Appendix B distinguishes their annotation designs. Training sizes follow experiment manifests; the separate agreement sample measures coding consistency and does not serve as a model test set.

Layer	Records	Support / composition	Role
Original corpus	22,840	Four source collections	Source corpus and original partition
Human reference	150	663 spans; 8 empty; 50 + 100 cohorts	Frozen evaluation targets
Original Silver-plus pool	2,451	100 reviewed; 730 conflict-origin; 1,621 other	Candidate labels; version-specific inclusion
Initial review	100	Versioned review layers (Appendix B)	Review of LLM labels; separate from evaluation
Post-generation review	100	Machine: 144 spans; reviewed: 148	Sampled machine–human quality check
JobBERT train / dev	2,156 / 169	Matched B1/B2 texts; different labels	Historical supervision and selection
Qwen B2 train / dev	2,150 / 169	3,646 / 345 spans; 869 / 82 empty	Earlier adapter fitting and selection
Additional dual-review QA	150	Both reviewers complete; 101 identical span sets	Quality check of the new Silver portion
Blinded agreement	50	Three independent coders; five + 15 training sentences excluded	Agreement analysis only; not a model test set
Expanded Silver (Codex)	9,540	8,842 / 348 / 350 train / validation / test	Adopted expanded supervision; additional Gold150 check

Expanded supervision and annotation profile. The adopted Codex pool contains 9,540 sentences from earlier supervision and additional cloud, public-sector, and listed-company collections. Selection prioritized requirement-related keywords. The supplementary 9,646-sentence pool differs in annotation channel and retained records. Appendix D records sampling and exclusions; Appendix C reports the two experiment-specific variants.

Two reviewers checked a further stratified sample of 150 newly annotated sentences, called QA150 in the released summaries. The sample contains 60 cloud, 45 public-sector, and 45 listed-company sentences and excludes the original 350 human-coded or reviewed sentences by identifier and normalized text. Both reviewers completed separate submissions with machine annotations visible. Historical human coding and review thus cover 500 unique sentences; the evaluation reference remains the original 150. The separate 50-sentence blinded agreement study is described below and is not added to this historical coverage count.

The supervision and reference subsets differ substantially in composition. Empty annotations occur in 40.4% of original Qwen training records and 48.5% of development records, compared

with 5.3% of reference sentences (Table 2). Occupational skills also dominate all five profiled layers (59.7–67.6%). These differences motivate per-type and empty-sentence diagnostics alongside micro-F1. Figure 3 shows length and type distributions; Appendix B records the medians.

Annotation Guidelines and Adjudication

The guidelines address three decisions that a list of skill terms cannot settle: whether a clause is eligible, how much text to retain, and which type fits its context. Annotation begins by determining whether a clause expresses a requirement for the candidate. Benefits, applicant-side procedures, company promotion, and pure outcomes do not qualify on their own; recruitment activities performed as occupational duties are assessed separately. Mixed clauses are considered separately. A sentence with no competency mention receives an empty annotation; insufficient context or an unresolved interpretation is recorded as uncertainty.

We adapted SkillSpan’s semantic-completeness and coordination principles for Chinese annotation (Zhang et al., 2022b). Our handbook asks annotators to select the shortest source span that preserves the complete requirement, without a fixed length cap. Generic proficiency triggers usually remain outside the span, but a modal can remain inside a complete colloquial capability expression. Necessary action heads and complements are retained. Coordinated activities are split only when each remains interpretable; a shared action and its objects stay together (Figure 4). Experience requires a separate check; generic work experience and duration alone are excluded. Explicit industry-background requirements use domain cores as broad-K proxies under the project convention. For occupational activities, the experience qualifier is omitted only if the remaining span preserves the requirement. It can stay outside coordinated items if each remains interpretable in context. Neither conjunctions nor experience qualifiers alone determine boundaries.

Type depends on what the clause requires: using SQL supports S, whereas knowledge of SQL principles supports K. A predicate can determine type without being part of the selected span. Familiarity alone does not decide type. General communication and coordination remain T inside duty clauses, whereas technical operations receive S. Language examinations receive L; educational and non-language qualifications receive K. Figure 4 shows original corpus examples with their archived annotations, and Appendix A gives further examples and rule sources.

Adjudication treats scope, boundaries, and type as separate decisions (Artstein and Poesio, 2008). There is no total priority order among L, K, S, and T. Under the current handbook, a mention with settled scope and boundaries but unresolved type may receive provisional S in the adjudication record, with alternatives logged. The handbook requires exclusion of the whole unresolved record from final training targets; provisional S is not an accepted target. Same-boundary alternatives, nested candidates, and unresolved overlaps are kept in the adjudication record rather than duplicated in the flat annotation layer. Appendix A gives the full rules and their history. Each completed experiment retains the labels assigned under its original handbook.

Human Annotation and LLM-Assisted Labeling

Human coding informed the handbook and finalized reference labels. Official OpenAI Codex generated separate Silver labels. Initial sample review preceded bulk annotation. Automated checks verified substrings, offsets, labels, and coverage; human review addressed interpretation.

Bulk annotation used the handbook available at generation time. Later amendments were recorded separately; completed experiments remained unchanged.

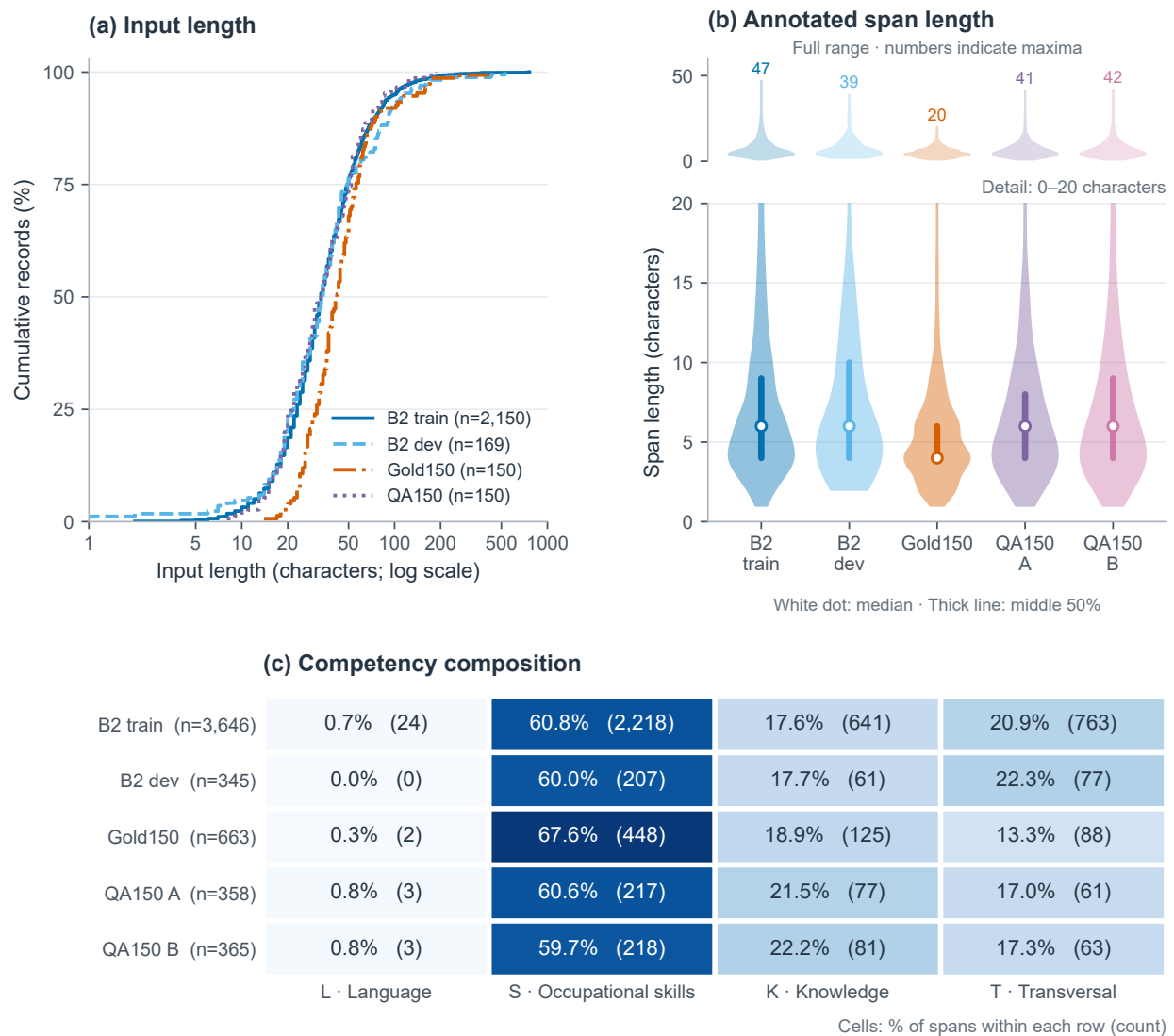


Figure 3. Benchmark annotation profile. (a) Empirical cumulative input-length distributions, including records with no spans; the horizontal axis is logarithmic. (b) Span-length violins: the main view enlarges 0–20 characters, covering 96.4–100.0% of spans in each layer; the upper overview retains the full range and labels the maxima. Both views use the same full-data density and equal maximum widths. White dots and thick lines mark medians and interquartile ranges. (c) Within-layer type percentages and span counts. B2 train/dev are the original 2,150/169-record Qwen manifests; Gold150 is the frozen 150-sentence reference. QA150 A/B are the two assisted-review layers of the same additional 150 sentences, shown once in (a). Lengths count source Unicode code points, including punctuation and spaces, and are not directly comparable with English token counts; repeated B2 records are retained. These historical supervision and review profiles exclude the independent three-annotator sample and the full expanded pool.

a) Language and transversal requirements

1802-s0008 | excerpt

4、良好的英文读写能力^L 5、优秀的沟通能力^T 和团队合作精神^T

Good English reading and writing, communication, and teamwork are distinct requirements.

b) Knowledge and programming skills

1806-s0005 | full sentence

2 深刻理解面向对象^K、熟悉设计模式^K、多线程和跨平台编程^S；

Understanding object orientation and design patterns differs from multithreaded and cross-platform programming.

c) Coordinated occupational actions

1803-s0002 | full sentence

3、独立操作竞价^S，如扩展长尾关键字^S，调整价格^S，优化曝光与转化率^S，并协助完成Listing优化^S；

Bid management includes keyword expansion, price adjustment, exposure/conversion optimization, and listing optimization.

d) Colloquial and interpersonal capabilities

1973-s0006 | excerpt

4、会聊天^T，有服务意识^T，善于与人沟通^T，具备较好的协调^T、沟通能力^T；

Conversation, service awareness, interpersonal communication, and coordination express transversal capabilities.

e) Experience requirements

1896-s0005 | full sentence

2、5年以上售前顾问^S或项目管理^S经验，有一定数量的大项目售前^S经验；

The record requests presales or project-management experience; generic duration and experience qualifiers lie outside the recorded spans.

Figure 4. Verbatim examples from the frozen human reference, with L/K/S/T spans and source IDs. Panels (a) and (d) are contiguous excerpts; the others show complete stored sentences. Highlights retain the archived v4.2.10 labels. English explanations are display aids, not model inputs; Appendix A gives current-handbook rules.

Codex supplied the adopted expanded labels; the alternative pool is supplementary. Unresolved records were excluded. Appendices A and D document versions and inclusion decisions.

Annotation Quality and Reference Freezing

Independent coding assesses handbook use without machine or peer suggestions; assisted review evaluates proposed labels. We report final-handbook agreement from the separate three-annotator blinded study (Table 3). Historical coding and review used different samples and handbook versions (Appendix B); their scores cannot track independent reliability over time.

Blinded coding protocol. Three annotators used handbook B.sop_v4.2.14 after familiarization and training on sets of five and 15 sentences, following the training-and-feedback approach discussed by Klie et al. (2024). They then independently coded the same random sample of 50 Silver teacher-pool sentences in separate Doccano projects, blinded to both machine suggestions and one another’s labels. Training sentences are excluded from formal agreement. The sample contains one sentence from each of 50 recorded advertisement identifiers; it assesses annotation agreement and is not a new model test set. Source counts and sampling-record limits appear in Appendix B. We computed agreement from the preserved pre-discussion exports, retaining all 50 sentences and all three coder pairs. Typed exact F1 measures agreement on complete spans without designating a coder as gold (Hripcsak and Rothschild, 2005); boundary-only F1 ignores type. Nominal Krippendorff’s α uses L/K/S/T/O assignments to all 2,111 source code points. Percentile 95% intervals use 10,000 source-stratified sentence-bootstrap samples, retaining all three annotations together. These intervals condition on the observed source mixture and coders.

Table 3. Independent blinded agreement among three annotators on 50 sentences. Character coefficients are Cohen’s κ unless marked α . Brackets give 95% bootstrap intervals for typed exact F1. “Equal” counts identical sentence-level span sets. Training samples are excluded.

Comparison	<i>n</i>	Character	Typed F1	Details
<i>Final-handbook independent blinded coding</i>				
A–B	50	0.800	0.657 [0.528, 0.767]	Boundary F1 0.762; equal 26/50
A–C	50	0.848	0.739 [0.612, 0.850]	Boundary F1 0.788; equal 33/50
B–C	50	0.803	0.594 [0.459, 0.717]	Boundary F1 0.721; equal 22/50
Mean / three-rater	50	α 0.817	0.663 [0.562, 0.759]	Mean boundary F1 0.757; all three equal 21/50

Independent blinded coding leaves substantial span-level disagreement. Mean typed F1 is 0.663, below the planned 0.80 project target, which is not a universal reliability cutoff. Boundary F1 is higher at 0.757 (95% CI [0.654, 0.846]), indicating additional type disagreement. Character-level $\alpha = 0.817$ (95% CI [0.755, 0.873]) measures a different unit and does not override these span disagreements (Artstein and Poesio, 2008). Of the 21 fully agreed sentences, ten are empty. The two L spans occur in one sentence, limiting category-level conclusions. The reproduction materials include frozen pre-discussion labels and diagnostics for this independent coding study.

Ethics and Data Use

The study analyzed recruitment advertisements without recruiting job applicants. Annotation and review were performed by members of the research team. Institutional ethics approval was not sought, and no formal institutional exemption is claimed. We cannot verify from the retained release records whether employer-contact names, telephone numbers, and email addresses were systematically audited or removed. A removal rate is therefore unavailable. The materials support academic research and peer review under the reuse terms in Data Availability.

EVALUATION PROTOCOL

The evaluation compares prompted extraction, measures gains from Silver-supervised Qwen adaptation, and tests encoder initialization under shared task-specific settings. Type, span-length, and empty-sentence diagnostics complement aggregate scores. Chinese systems share the human reference; Silver-target and external-task evaluations retain separate datasets and protocols.

Datasets and Splits

The original supervision experiments train and select models on Silver data and score them against the fixed human reference. JobBERT uses 2,156 training and 169 development records. Six training records match three development texts after NFC normalization; removing them gives Qwen’s 2,150/169 split. Repeated texts remain within training. Neither split contains reference identifiers or complete NFC-normalized reference sentences. Partitioning is at sentence level, so different sentences from one advertisement may occur in different splits. Scoring preserves the original sentence text and its character offsets for matching predicted and reference spans. Supplementary expansion experiments use 8,842/348/350 training, validation, and test sentences from the adopted Codex pool, and 8,943/351/352 from the alternative pool. Both combine earlier supervision with additional annotations, but differ in annotation channel and the examples retained after checking. Records retained after the channel replacement keep their split assignments. For Qwen, Silver-validation exact F1 selects the checkpoint, Silver-test F1 measures agreement with held-out generated labels, and the human reference provides an additional evaluation. Reference sentences are excluded from gradients and checkpoint selection. Variants and results are in Appendix C; split checks are in Appendix D.

Evaluation Metrics

To distinguish exact extraction from approximate boundary agreement, the primary metric is typed exact-span micro-F1: a prediction matches only when both character boundaries and the LSKT type agree with the reference. Typed relaxed micro-F1 uses one-to-one greedy matching of same-type spans with intersection-over-union (IoU) of at least 0.5:

$$\text{IoU}(u, v) = \frac{|u \cap v|}{|u \cup v|}, \quad \text{IoU}(u, v) \geq 0.5. \quad (1)$$

Here, u and v are the sets of character positions covered by a predicted and a reference span, and $|\cdot|$ counts positions. For either matching rule, TP counts matched pairs, FP unmatched predictions, and FN unmatched reference spans, summed across sentences:

$$P_{\text{micro}} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad R_{\text{micro}} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad F_{1,\text{micro}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}. \quad (2)$$

All 150 human-reference sentences are scored. Expanded Qwen experiments also report scores on their Silver validation and test sets. Partially rejected responses retain accepted spans; malformed responses or responses with no accepted spans are scored as empty predictions. Empty-empty sentences add no span counts. Appendix C summarizes parser decisions and matching order and links the full scorer specification. Boundary exact micro-F1 matches span boundaries

after collapsing types. Per-type scores retain the original L/K/S/T labels and include all reference sentences, including sentences without spans of the evaluated type. Across three training seeds, we report the mean and sample standard deviation:

$$\bar{f} = \frac{1}{n} \sum_{i=1}^n f_i, \quad s_f = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (f_i - \bar{f})^2}, \quad n = 3. \quad (3)$$

Inference-only and expanded-pool results are single runs unless otherwise stated. The original-task reruns report their own seed counts. Appendix C documents resampling procedures.

Models and Baselines

JobBERT-zh provides a sequence-labeling baseline with a Chinese recruitment-domain-pretrained encoder and a CRF head. We compare Qwen2.5-14B-Instruct (Qwen) (Qwen Team, 2025) with and without a project LoRA adapter (Hu et al., 2022) for generative extraction. The shared-guideline inference comparison includes GPT (API), Claude (API; Sonnet 4.5), DeepSeek-V4-Pro (DeepSeek), Kimi k2.6 (Kimi), and Qwen. DeepSeek is evaluated with thinking enabled and disabled, giving six configurations. Appendix C records service identifiers, access routes, and inference settings.

Implementation Details

Supervision and model selection. JobBERT conditions start from the same released encoder and CRF. Earlier supervision (B1 in the experiment files) uses the historical extraction labels; revised supervision (B2) combines updated guidance, LLM annotation, and adjudication. The conditions share training and development texts, with label changes in 1,440 training records and 107 development records. Development typed exact F1 selects checkpoints, so this comparison changes training supervision and selection targets together.

Qwen starts from its pretrained Instruct weights and fits a fresh LoRA adapter for each of seeds 42, 43, and 44 on the revised 2,150/169 split. Training uses a fixed system instruction, user template, and five schematic examples. Targets contain the literal span text, occurrence index, and type. Loss applies to the assistant JSON and closing turn token. Training uses BF16, and development exact F1 selects the adapter. Table 4 gives the remaining settings. Historical JobBERT implementation gaps are documented in Appendix D.

Table 4. Original JobBERT and Qwen training settings. Both select checkpoints by development typed exact F1; reproduction notes retain implementation and selection records.

Setting	JobBERT-zh B1/B2	Qwen LoRA
Learning rate	2×10^{-5}	10^{-4}
Batch size	16	1; 16-step gradient accumulation
Maximum input length	256 tokens	8,192 tokens
Training limit	6 epochs; patience 2	6 epochs; patience 2

Qwen uses LoRA with rank 16, scaling 32, and dropout 0.05, applied to the attention q/k/v/o projections.

Inference. Each request contains one sentence’s identifier and text, together with the shared instruction and five fixed examples. A parser converts occurrence-based output into character spans. Qwen baseline and adapter inference use the same base checkpoint, prompts, parser, BF16 precision, deterministic decoding, and 2,048-token output cap. DeepSeek uses caps of 2,048 tokens without thinking and 8,192 with thinking. Appendix C lists service settings and output checks.

External Benchmarks and Multilingual Encoders

To test whether domain-pretrained initialization helps consistently across recruitment tasks, the experiments compare XLM-R-large (Conneau et al., 2020) and ESCOXLM-R (Zhang et al., 2023). They use six public datasets: SkillSpan (Zhang et al., 2022b), Kompetenzer (Zhang et al., 2022a), German ICT (Gnehm et al., 2022b), Green (Green et al., 2022), Sayfullina (Sayfullina et al., 2018), and FIJO (Beauchemin et al., 2022). Separate skill and knowledge streams in SkillSpan and Kompetenzer give eight extraction tasks, each retaining its original entity types.

Both encoders use fresh linear first-subword BIO heads, AdamW with a learning rate of 2×10^{-5} , batch size 8, and a maximum of 20 epochs. Training uses seeds 42, 43, and 44, with development typed exact token-span micro-F1 selecting checkpoints. For the Chinese comparison, both encoders and their new nine-label BIO heads were fine-tuned on B2’s 2,150 training and 169 development records. Both models process the original Chinese characters directly. The new BIO head labels the first subword of each character; predictions from chunks of at most 128 characters are joined, including the remainder of long sentences. Tokenizer placeholders preserve positions for whitespace and otherwise discarded characters; scoring uses the original text and offsets. Development F1 selects the checkpoint. Gold150 is used only for diagnostic scoring within these runs, although it was previously used in guideline development. Exact text and identifier checks found no overlap across the three partitions; advertisement-level independence is not established.

The Chinese comparison uses the same optimizer settings, maximum epochs, and three seeds as above. Its linear heads and training budget differ from JobBERT’s CRF and Qwen LoRA, so shared-reference scores support descriptive comparisons rather than a controlled ranking of those architectures. We report typed exact and relaxed character-span F1, boundary-only F1, and macro-F1 over L/K/S/T. L has only two reference spans but receives one quarter of macro-F1 weight.

External-task paired sentence-bootstrap intervals estimate the mean ESCOXLM-R minus XLM-R difference for three fixed trained seeds (2,000 resamples; seed 20260921). We report training-seed SD separately. Intervals omit advertisement clustering and multiple-comparison adjustment.

Appendix E gives separate protocols for five-seed MaChAmp-CRF reruns on SkillSpan and German ICT, five-seed Kompetenzer classification, and single-seed NNOSE retrieval. These preserve their task-specific scoring: Danish BIO extraction differs from fine-grained classification of supplied spans, and the public German ICT task (Gnehm et al., 2022b) is distinct from the EDU/EXP/LNG extraction and taxonomy-classification pipeline (Gnehm et al., 2022a). Languages and annotation schemes differ, so absolute scores do not rank extraction difficulty.

The Chinese encoder comparison uses 10,000 draws (seed 20260924), jointly resampling the same 150 sentences for both models and all three fixed seeds. Each draw averages paired differences in recomputed per-seed micro-F1. This post hoc interval describes reference-sentence sampling uncertainty, not population generalization, and omits advertisement clustering.

RESULTS AND ANALYSIS

Inference on the Human Reference

GPT (API) achieved the highest typed exact micro-F1 among the six configurations, followed by DeepSeek with thinking disabled (Table 5). Their paired difference is 0.020, with a 95% sentence-bootstrap interval of $[-0.026, 0.064]$ (Table 6). The interval includes zero, so the point scores do not establish a stable ordering. These are comparisons of the recorded configurations, which use shared instructions but different inference budgets.

Table 5. Shared-guideline inference configurations on the human reference ($n = 150$). P/R are typed exact precision/recall. Each configuration is a single run. Inference budgets are configuration-specific. Qwen uses no project adapter.

Recorded configuration	P	R	Exact F1	Relaxed F1
GPT (API)	0.676	0.658	0.667	0.766
DeepSeek (non-thinking)	0.651	0.643	0.647	0.761
Claude (API)	0.598	0.597	0.598	0.709
DeepSeek (thinking)	0.713	0.505	0.591	0.671
Kimi	0.581	0.563	0.572	0.714
Qwen	0.507	0.281	0.361	0.470

Both DeepSeek rows use the same model. Non-thinking and thinking denote inference modes, with output caps of 2,048 and 8,192 tokens respectively. Table 14 lists the recorded identifiers, access routes, and settings. Cohort scores and detailed output-status counts accompany the reproduction materials.

Table 6. Paired typed exact-F1 differences on Gold150, with conditional 95% sentence-bootstrap intervals. All failures remain in the scored predictions.

Comparison	Difference	95% interval
GPT (API) minus DeepSeek non-thinking	+0.020	$[-0.026, 0.064]$
Qwen LoRA 42 minus no adapter	+0.138	$[0.071, 0.208]$
Qwen LoRA 43 minus no adapter	+0.202	$[0.132, 0.276]$
Qwen LoRA 44 minus no adapter	+0.197	$[0.127, 0.271]$
Qwen LoRA seed mean minus no adapter	+0.179	$[0.113, 0.249]$

Fine-Tuning with Silver-plus Labels

All three Qwen adapters improved exact precision and recall over the no-adapter baseline under the same inference protocol (Table 7, panel A). Mean typed exact micro-F1 increased from 0.361 to 0.540 ± 0.035 . Each seed's paired interval against the baseline also excludes zero (Table 6), conditional on the recorded runs and reference sentences. Under the same parser, partially-rejected responses fall from 69 without an adapter to 2, 4, and 1 after adaptation. These gains do not separate semantic extraction from output-format learning.

Table 7. Evaluation of Silver-supervised fine-tuning on Gold150 ($n = 150$). F1 scores (0–1) are mean $\pm$ sample SD over seeds 42, 43, and 44; Qwen without an adapter is a single run.

A. Qwen: fixed inference protocol				
Condition	Train / dev	Typed exact	Typed relaxed	Boundary exact
No adapter	—	0.361	0.470	0.414
B2 LoRA	2,150 / 169	0.540 ± 0.035	0.662 ± 0.033	0.571 ± 0.038
B. JobBERT-zh: historical supervision and selection				
Supervision and selection labels	Train / dev	Typed exact		
Earlier (B1)	2,156 / 169	0.142 ± 0.014		
Revised (B2)	2,156 / 169	0.554 ± 0.005		

A: same base checkpoint, frozen messages, occurrence schema, parser, and decoding; adapters selected by B2 development F1. Boundary exact matches spans with types collapsed. B: same initial encoder/CRF and matched training/development texts, with both label sets changed. Limitations and Responsible Use covers split and checkpoint limitations; the run ledger gives per-seed Qwen scores for all three training runs.

The historical JobBERT configuration achieved higher reference agreement with revised supervision (Table 7, panel B). The two conditions start from the same encoder and CRF and use matched training and development texts, but revise both sets of labels. Their difference therefore reflects changes to supervision and checkpoint selection together.

Supplementary Analyses

The Qwen improvements extend across the three reported span-length bins: every adapter increases exact recall in each bin (Figure 5). False-positive spans on the eight empty-reference sentences also decrease, although the changes vary by seed. The figure shows individual-seed results; Table 8, panel B, compares aggregated category scores across models. Exploratory experiments with expanded supervision appear in Appendix C. Those runs change the source composition, annotation targets, and retained examples together. They do not isolate an expansion effect. Available diagnostics do not explain the lower expanded JobBERT scores.

Chinese Model Diagnostics and External Comparisons

The Chinese comparison does not establish a reliable advantage of ESCO initialization. The encoders achieved mean exact F1 of 0.522 (XLM-R-large) and 0.538 (ESCOXLM-R), with all six prediction files independently rescored against the original reference (Table 8, panel A). The mean difference was 0.016; its paired 95% interval was $[-0.003, 0.034]$. Four-type macro-F1 is 0.607 and 0.622; excluding the two-span L class gives descriptive K/S/T macro values of 0.542 and 0.562, respectively. This check does not replace the four-type metric. Silver-supervised Qwen adaptation improved both S and K extraction: mean typed exact F1 increased by 0.148 and 0.210, respectively (Table 8, panel B). Each adapter improved K, S, and T relative to the no-adapter run (Figure 5). Qwen has the highest S mean and ESCOXLM-R the highest K mean among these three adapted models. These cross-model comparisons are descriptive because prediction heads, training budgets, and inference procedures differ. Archived runs show mixed effects of domain-pretrained initialization across external tasks (Table 9). ESCOXLM-R had a higher mean typed exact F1 on SkillSpan skills and Sayfullina and a

lower mean on the other six streams. The largest decreases were on SkillSpan knowledge and German ICT; only these two pointwise bootstrap intervals excluded zero (Appendix E, Table 18). All seeds were retained, including early-selected runs. FIJO has only 50 test sentences. The original-task reruns answer related but separate questions. Combining English and Danish training improves Danish classification macro-F1 for both tested encoders. NNOSE retrieval improves the single-seed Green results but not SkillSpan or Sayfullina. Appendix E separates classification, retrieval, and extraction protocols for these and public-subset MaChAmp reruns.

Table 8. Chinese model diagnostics on Gold150 (0–1). Results across three training seeds are mean $\pm$ sample SD; Qwen without an adapter is a single run.

A. Chinese encoders: overall F1				
Encoder	Typed exact	Typed relaxed	Boundary exact	Macro-F1
XLM-R-large	0.522 $\pm$ 0.022	0.650 $\pm$ 0.021	0.571 $\pm$ 0.022	0.607 $\pm$ 0.016
ESCOXLM-R	0.538 $\pm$ 0.021	0.662 $\pm$ 0.022	0.582 $\pm$ 0.024	0.622 $\pm$ 0.010

B. Typed exact F1 by competency type				
Model	S ($n = 448$)	K ($n = 125$)	T ($n = 88$)	L ($n = 2$)
Qwen, no adapter	0.377	0.302	0.380	0.000
Qwen B2 LoRA	0.525 $\pm$ 0.039	0.512 $\pm$ 0.063	0.638 $\pm$ 0.017	0.800
XLM-R-large	0.506 $\pm$ 0.024	0.495 $\pm$ 0.037	0.627 $\pm$ 0.006	0.800
ESCOXLM-R	0.518 $\pm$ 0.029	0.537 $\pm$ 0.009	0.631 $\pm$ 0.020	0.800

All adapted models use the B2 split (2,150/169 training/development records) and seeds 42/43/44. Encoders use fresh linear LSKT heads; Qwen uses a LoRA adapter and the fixed inference protocol in Table 7. All four types are predicted jointly. Boundary exact collapses types; macro-F1 weights L/K/S/T equally. For L, each trained run has 2 TP, 1 FP, and 0 FN; Qwen without an adapter has 0 TP, 1 FP, and 2 FN. L seed SD is omitted because two reference spans cannot establish reliability. Gold150 informed guideline development and is not a new blind test.

Table 9. Common-protocol typed exact token-span micro-F1 (0–1). Each row reports mean $\pm$ sample SD across three training seeds for one dataset and annotation stream.

Dataset	Stream	Train/dev/test	XLM-R-large	ESCOXLM-R
SkillSpan	Skill	4,800/3,174/3,569	0.567 $\pm$ 0.012	0.577 $\pm$ 0.006
SkillSpan	Knowledge	4,800/3,174/3,569	0.698 $\pm$ 0.021	0.672 $\pm$ 0.052
Kompetencer	Skill	778/346/262	0.545 $\pm$ 0.028	0.527 $\pm$ 0.004
Kompetencer	Knowledge	778/346/262	0.618 $\pm$ 0.035	0.593 $\pm$ 0.044
Gnehm ICT	ICT	19,889/2,332/2,557	0.891 $\pm$ 0.005	0.867 $\pm$ 0.032
Green	Native types	8,669/964/335	0.493 $\pm$ 0.009	0.492 $\pm$ 0.007
Sayfullina	Skill	3,705/1,855/1,851	0.920 $\pm$ 0.004	0.925 $\pm$ 0.005
FIJO	Native types	399/50/50	0.343 $\pm$ 0.019	0.339 $\pm$ 0.007

Both encoders use fresh linear BIO heads, 20 epochs, and development-set checkpoint selection. SkillSpan and Kompetencer have separate skill and knowledge streams; Green and FIJO retain their entity types. German data use the public ICT release. The shared protocol differs from the original CRF and NNOSE methods.

Boundary and Type Disagreements in Human Coding

The three-annotator study shows that character agreement alone understates the difficulty of reproducing complete typed spans (Table 3). Boundary-only agreement exceeds typed agreement, so disagreements concern both extent and type. Historical calibration cases explain why the handbook distinguishes job-use tools from knowledge and general communication from technical operations; they do not establish the distribution of errors in the separate blinded sample.

Implications for Benchmark Use

The no-adaptor Qwen score rises from 0.361 under exact matching to 0.470 under overlap-tolerant matching (Table 5). This difference shows how boundary tolerance affects the evaluation. The prediction–reference examples in Appendix C illustrate boundary and type mismatches, but have not been adjudicated into a linguistic error taxonomy.

Agreement with generated labels does not by itself establish agreement with human annotations. The Qwen run using the adopted expanded Codex pool achieves 0.809 on its Silver test and 0.585 on the human reference (Table 15). The two sets differ in both texts and targets, so the gap cannot estimate Silver-label error, but it illustrates why both evaluations are needed.

Matched Qwen runs provide the clearest adaptation evidence because inference is fixed. JobBERT and expansion studies jointly change labels, selection, or source composition, so they cannot isolate those effects. Type, length, and empty-sentence diagnostics qualify aggregate scores.

LIMITATIONS AND RESPONSIBLE USE

Reference design and coverage. The calibration/challenge reference informed guideline development; exclusion from gradients and checkpoint selection does not remove that dependence. Its small, uneven type distribution and keyword-prioritized expansion limit generalization. Independent blinded coding yields mean typed F1 of 0.663, with only two L spans. The 50-sentence teacher-pool sample does not establish corpus-wide reliability, and historical assisted review is a different design. Annotations describe advertised requirements, not applicants’ abilities.

Data access and reproducibility. Public labels and predictions support inspection and rescoring, but the complete expanded-pool training inputs remain unavailable. Collection records, historical implementation details, and label-version compatibility checks are incomplete, limiting retraining and verification. Component-level access and reuse terms are specified below. Further applications require source permissions, privacy safeguards, human oversight, and external validation.

Experimental comparisons. Sentence-level splits, residual JobBERT train/development matches, and incomplete CRF selection history limit independence claims. Expanded-pool studies use single seeds; inference budgets and access configurations differ. Table 14 records model-provenance limits. External tasks differ in labels, splits, scorers, and selection. Bootstrap intervals condition on trained seeds without advertisement clustering or multiple-comparison adjustment.

CONCLUSION

Chinese-SkillSpan provides a benchmark for delimiting and typing competency mentions in Chinese job advertisements, with explicit treatment of coordination, contextual types, and negative cases. It links original text and character offsets to an ESCO-informed handbook and separates human-reference evaluation from Silver supervision. Independent coding by three annotators,

blinded to machine and peer labels, measures final-handbook consistency separately from historical coding and assisted review. Its remaining boundary and type disagreements show where adjudication and rule clarification are still needed.

On the development-used human reference, Qwen adaptation improved end-to-end extraction under fixed inference rules. Released Chinese-supervised encoder checkpoints and predictions permit inference checks and rescoring; their paired interval does not establish an ESCO initialization advantage. Boundary-sensitive scores and the type, length, and empty-sentence diagnostics identify where further assessment is needed. These findings support diagnostic use of the benchmark and its annotation resources. Blinded agreement assesses annotation consistency, not model generalization to new advertisements or compatibility across historical label versions.

DATA AVAILABILITY

The dataset archive (v0.1.3; DOI: 10.5281/zenodo.22698504) contains the corpus, handbook, 150-sentence human reference, and Silver-plus B2 training/development files; 10.5281/zenodo.22288337 is the concept DOI. The historical annotation-layer release separately provides pseudonymized coding and QA150 exports with original text and offsets. The independent blinded three-annotator study provides frozen labels, source identifiers, agreement code, and results. These later annotation studies are outside the original Zenodo archive.

The Qwen reproduction archive (DOI: 10.5281/zenodo.22851581) contains three B2 LoRA adapters, selected expanded-Silver checkpoints, predictions, configurations, and scoring tools. Complete texts and labels for the adopted 9,540-record Codex pool and the supplementary 9,646-record alternative remain unavailable for public retraining. The six Chinese-supervised XLM-R/ESCOXLM-R checkpoints are linked in the Hugging Face model index; their code, configurations, predictions, and verification records are in the audit archive (DOI: 10.5281/zenodo.22937245). The evidence index separates local rescoring of archived predictions from server-reported checkpoint inference.

The JobBERT-zh initialization supplies the domain-adapted encoder and inherited CRF; the B2 continuation supplies continued checkpoints, with seed 42 as its default. The project repository and reproduction guide link code, preprocessing, scoring, and experiment records. Table 16 summarizes access and verification; Appendix E links external-benchmark protocols and results. Reuse terms vary by component. Explicitly covered software and documentation retain Apache-2.0. Author-controlled annotations and research materials permit academic research and peer review with attribution; this permission excludes third-party recruitment text. Model checkpoints and adapters follow their accompanying notices and base-model licences. The data-access guide specifies these terms; Appendix D records earlier versions and access limits.

APPENDIX A. CODING GUIDE AND EXAMPLES

This guide follows Handbook v4.2.14 and complements Figure 4. The reader-facing v4.2.14 edition clarifies examples for interpretation of the current rules; unresolved cases retain that status. This documentation does not replace the studies' frozen handbooks or labels. The source package contains the bilingual manuals and crosswalk; Section A.4 records version boundaries.

A.1 Label definitions and context tests

Square brackets mark selected source substrings, with a separate type code after each span. Brackets, colours, and type codes are display aids and are not source characters; boundaries remain identifiable in grayscale. Examples illustrate the handbook, not an additional evaluation sample. *Source key.* E: ESCO's skills hierarchy and transversal concepts (European Commission, 2024); Q: EQF definitions (Council of the European Union, 2017); Z: related annotation guidance in SkillSpan, Appendix B (Zhang et al., 2022b); I: unitization and categorization in agreement analysis (Artstein and Poesio, 2008); P: our V4.2.14 operational decision. Combined marks indicate adaptation, not verbatim rules; Chinese examples and exact boundaries are project illustrations.

Table 10. Current LSKT decisions: definitions, examples, and contextual contrasts.

Label	Operational definition	Chinese example
L: Language	Natural-language proficiency, language names, levels, examinations, and certificates. ^[E,P]	[英语六级] [L]; [日语 N2] [L]
K: Knowledge	Facts, principles, theory, and fields of knowledge. ^[Q] Education and non-language qualifications map to K by project convention. ^[P]	[计算机专业] [K]; [本科及以上学历] [K]
S: Occupational skill	Application of a tool or method, or an occupational activity indicated by the context. ^[Q,P]	使用 [SQL] [S]; [数据分析] [S]
T: Transversal competence	General communication, cooperation, cognition, self-management, and dispositions. Duties may also express T. ^[E,P]	[客户汇报] [T]; [责任心] [T]
Context test	Use versus principles distinguishes S from K; general coordination differs from a technical operation. No fixed label priority applies. ^[Q,E,P]	了解 [SQL 原理] [K]; [协调外部资源] [T]; [接口对接] [S]

A.2 General annotation decisions

G1–G15 are quick-reference identifiers. Their source sections and amendment IDs (R08–R23) are mapped in the release; they do not replace the handbook's rule numbering.

Table 11. Annotation principles for target identity and complete source boundaries.

Rule	Annotation principle	Chinese example
G1	The annotation scheme distinguishes L, K, S, and T. Collapsed categories are used only in secondary evaluation. ^[E,P]	[英语六级] [L]; [计算机专业] [K]; [数据分析] [S]; [责任心] [T]
G2	Final annotations contain non-overlapping spans. A sentence may contain several independent mentions, with each character assigned to at most one span. ^[P]	熟悉使用 [Word] [S], [Excel] [S], [PPT] [S] 等办公软件 No additional covering list span.

Table 11 (continued)

Rule	Annotation principle	Chinese example
G3	Each mention is a contiguous source substring that preserves spelling and internal spaces. Any normalization is stored separately. [Z,P]	熟悉 [Jekins] [S] Keep the source spelling, not Jenkins.
G4	Annotators select the shortest span that expresses a complete requirement. Words and identifiers remain intact; completeness takes priority over length, with no fixed length cap. [Z,P]	[支持服务] [S] Not the fragment 支持服.
G5	Generic proficiency triggers usually remain outside the span, while heads needed to express the requirement remain inside. Excluded triggers can still inform type assignment. [Z,P]	会使用 [Excel] [S]; [会聊天] [T]; 有 [服务意识] [T]
G6	Coordinated items are separated when each remains meaningful in context. Shared qualifiers may stay outside the spans, and implied words are not added. [Z,P]	熟悉使用 [Word] [S], [Excel] [S], [PPT] [S] 等办公软件 Shared-experience case: see A.3.
G7	A necessary action, object, or shared head remains within the span when splitting would change its meaning. Span length alone does not determine type. [Z,P]	[优化曝光与转化率] [S] Isolated metric names would lose the optimization activity.

Table 12. Annotation principles for scope, experience, type, and unresolved records.

Rule	Annotation principle	Chinese example
G8	Benefits, company promotion, administrative instructions, experience durations, and pure outcomes are outside competency scope on their own. Mixed clauses are assessed separately. [Z,P]	五险一金, 带薪年假 → []; [分析推广效果] [S], 保证流量/用户增长
G9	Experience wording is omitted only when the remaining activity preserves the requirement. Necessary practical-experience wording is retained. Explicit industry background follows the handbook's broad-K convention. Experience wording is never added to the source. [P]	[机器人竞赛经验] [S] Retain the suffix if a bare competition name changes the requirement; do not label generic 经验 alone.
G10	Education qualifications include their threshold wording. Language examinations and certificates receive L; non-language qualifications receive K. [Z,P]	[本科及以上学历] [K]; [大学英语 6 级] [L]; 了解 [ISO 27001] [K] Standard familiarity, not a claim of personal certification.

Table 12 (continued)

Rule	Annotation principle	Chinese example
G11	Tool and method mentions are typed from their predicate context. Occupational use supports S; principles or theory support a complete K phrase. Familiarity alone does not determine type. [Q,P]	使用 [SQL] [S]; 了解 [SQL 原理] [K]
G12	Occupational functions are distinguished from general cognition, communication, and coordination. A duty clause may contain T; usefulness across industries alone does not establish T. [E,P]	[编码能力] [S]; [沟通能力] [T]; [协调外部资源] [T]
G13	Scope, boundaries, and type are resolved separately, without a total label priority. Once scope and boundaries are valid, type-only uncertainty may receive provisional S with alternatives logged; this is excluded from final training targets. [I,P]	Heading alone: 能力提升 Insufficient scope evidence: record as unresolved, not a confirmed empty negative.
G14	Offsets count original Unicode code points, beginning at zero and excluding the end position. The indexed substring must reproduce the mention exactly; segmentation is a derived view. It does not replace the fixed human-reference offsets. [P]	text[start:end] == span Keep the original text, internal spaces, and offsets consistent.
G15	Alternatives and overlaps are recorded separately from the final flat layer. An unresolved record is excluded in full from training, with confirmed empty, missing, and unresolved annotations distinguished. [P]	五险一金 → confirmed []; 能力提升 without context → unresolved. Known partial spans do not make an unresolved record training-ready.

A.3 Shared experience: split only when meaning survives

The V4.2.14 amendment (R23) clarifies a shared-experience construction. [P] Each item remains identifiable in context; exclude the shared experience qualifier and do not add a covering span.

4、三年以上大中型 [互联网公司经营] [S] 或 [BI] [S] 相关工作经验, [互联网公司广告运营分析] [S] 优先;

Three or more years of operations or BI experience in large or medium-sized internet companies; internet-company advertising operations analysis preferred.

Selected offsets: [9, 16), [17, 19), and [26, 37). BI is S in this context, not by a universal dictionary rule. In contrast, G7 retains the shared action in 优化曝光与转化率 (*optimize exposure and conversion rate*); splitting must not leave bare outcomes. The earlier long span in the initial reviewed LLM-annotation dataset remains historical rather than being silently overwritten.

A.4 Version boundaries

The human-reference metadata identify v4.2.10, while the original bulk-generation prompt follows v4.2.12. The current handbook is v4.2.14. Later amendments do not retrospectively change completed annotations. Inspection confirmed v4.2.10 metadata on all 150 reference records; the B2 rows do not carry individual handbook-version fields. One documented difference is present in B2 training record 1838-s0008: its historical two-span annotation retains a shared experience

phrase that R23 later splits into three spans. This named record is absent from Gold150 and B2 development. A rule-family screen covers all 2,469 reference, training, and development records, but flags candidates for contextual review rather than proving semantic compatibility. The original targets and scores remain unchanged. Figure 4 reproduces frozen v4.2.10 human-reference annotations, whereas Tables 10–12 illustrate decisions under the current v4.2.14 handbook. Later rule amendments do not retrospectively alter the historical annotations or reported scores. The version ledger records experiment-specific protocols, identifiers, hashes, and unresolved bindings. The accompanying manuals preserve original and revised spans and their rationale. Human-reviewed Silver-plus remains supervision unless separately designated and isolated for evaluation. Evaluation uses the frozen reference, shared protocol, occurrence schema, parser, and scorer; the reproduction guide maps these terms to their released files (Appendix D).

APPENDIX B. ANNOTATION-QUALITY EVIDENCE

Historical coding and assisted-review comparisons below use designs distinct from the independent blinded study in Table 3. The initial 50-sentence calibration study involved independent raw-text coding before adjudication under the historical handbook. The sentences were sampled from the 980-record conflict queue. Available exports reproduce 98 exact matches between 211 and 230 spans, giving F1 0.444. Character-level κ is 0.567 for L/K/S/T plus O, over all 2,571 source code points, including spaces and punctuation. All 50 identifiers belong to the calibration portion of Gold150 and none occurs in the B2 training or development identifiers. Thus, the study supplies independent-coding evidence, but is not an additional test cohort. These calculations use the available later exports; exact correspondence with the original frozen files has not been verified. The calculation record preserves the arithmetic and supporting counts. The reference’s challenge cohort contributes 415 spans (290 S, 83 K, 41 T, one L), and the calibration cohort contributes 248 (158 S, 42 K, 47 T, one L). Seven challenge sentences and one calibration sentence have no annotated span. The identifiers contain 61 document prefixes; these are identifier-based groups rather than independently reconstructed advertisements. The adjudicated reference remains separate from the original coding and assisted-review layers. Character-level Cohen’s κ adjusts observed agreement for agreement expected by chance from the two coders’ marginal label frequencies across the source character positions:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \quad p_e = \sum_{c \in \mathcal{C}} p_A(c) p_B(c). \quad (4)$$

Here, p_o is the fraction of source character positions with identical labels, and $p_A(c)$ and $p_B(c)$ are the two coders’ label proportions. The label set $\mathcal{C}$ comprises L/K/S/T and O. For the four-label scheme plus O, the available exports yield $p_o = 0.771$ and $p_e = 0.471$, giving $\kappa = 0.567$. These character-level quantities are distinct from the span counts in Eq. (2); the 2,571 positions are nested within 50 sentences, not 2,571 independent samples.

Empty–empty sentences contribute no span true positives. Span matching and character-level agreement use different units; reproduction materials document optional category projections. The post-generation review sample instead compares a machine-prefilled layer with human review: 140 exact typed matches among 144 machine and 148 reviewed spans give P/R/F1 of 0.972/0.946/0.959. Ninety-five sentences have identical complete span sets. The sample was stratified by historical source, empty/nonempty status, and text length from 2,179 eligible records; two

small strata totalling 11 records were unsampled. The estimate describes this reviewed sample; it does not provide an unbiased estimate of accuracy across the corpus.

The 15-sentence auditor subset is nested within the post-generation Silver review cohort. Both annotators had access to machine coding suggestions. Their human layers contain the same 27 spans, yielding exact span F1 and character kappa of 1.000. These values describe agreement under machine-assisted review, where shared suggestions can influence both annotators; they are not evidence of independent blind coding or corpus-wide reliability under the final handbook.

Historical annotation profile. Median input lengths are 33 Unicode code points in each B2 supervision split, 41.5 in the reference, and 33.5 in the additional review sample. Median span length is six code points in supervision and both review layers, compared with four in the reference. Counts include spaces and punctuation; Figure 3 shows these historical distributions.

Dual-Reviewer QA on the Expanded Silver Sample

The completed export contains both human layers for the 150 preselected sentences, with all confirmations recorded separately for each reviewer. Identifiers exactly match the stratified draw, and normalized sentence texts have no overlap with the original 350. The sample includes 60 cloud, 45 public-sector, and 45 listed-company sentences. Machine suggestions were visible, so submissions measure assisted reviewer–reviewer agreement rather than blind agreement.

Table 13. Paired agreement in the additional QA150 sample. Exact typed span F1 uses original pre-adjudication span sets. Percentile 95% confidence intervals use 10,000 sentence resamples within source strata (seed 20260917), conditional on the source quotas.

Comparison	Exact F1	95% interval	Identical sentences
Reviewer A–Reviewer B	0.816	[0.769, 0.860]	101/150
Astra–Grok suggestions	0.962	[0.937, 0.983]	138/150
Reviewer A–Astra	0.823	[0.776, 0.866]	100/150
Reviewer B–Astra	0.850	[0.803, 0.892]	108/150

The 49 sentences with differing human span sets remain part of the raw agreement calculation. The exported annotations are not replaced by their intersection or a consensus layer. Character-level calculations assign O/L/K/S/T to original Unicode-code-point positions, including spaces and punctuation. Two positions have conflicting types within one reviewer layer and are excluded from both character sequences, leaving $N = 5,990$ positions. All original spans remain in span-level F1. These raw review layers are not substituted for Gold150; the affected record is identified in the accompanying calculation record. We have not verified final adjudication of these disagreements or traced the resulting decisions into each later training target. The reported coefficients therefore describe the preserved raw review layers. The resulting observed agreement over retained character positions is $p_o = 0.934$, with $p_e = 0.439$ and $\kappa = 0.883$.

For nominal Krippendorff’s α , let m_c count assignments to category c across both reviewers, with $\sum_c m_c = 2N$. The observed and expected disagreements are

$$\alpha = 1 - \frac{D_o}{D_e}, \quad D_o = 1 - p_o, \quad D_e = 1 - \frac{\sum_c m_c(m_c - 1)}{2N(2N - 1)}. \quad (5)$$

Here $D_o = 0.066$ and $D_e = 0.560$, giving $\alpha = 0.883$. The character coefficients and span F1 describe different units of agreement (Artstein and Poesio, 2008); the many O positions contribute

to the character calculation. Agreement between two machine suggestion layers likewise measures concordance, not accuracy against an independent human reference. Export hashes, calculation code, and detailed diagnostics accompany the reproduction record.

Independent blinded coding by three annotators. The formal sample contains eight listed-company, 16 public-institution, and 26 Tianchi sentences. We used random sampling to select the formal sentences. Each annotator completed familiarization and training before independent coding. The archived exports establish sample membership and source counts, but the complete candidate frame and sampling execution log were not included in this release. The original draw therefore cannot be independently reproduced from the release; source-stratified bootstrap resampling quantifies agreement uncertainty conditional on this sample. Five- and 15-sentence training sets are excluded from formal agreement. The analysis uses pre-discussion labels, not later edited exports. The October 2 reader-edition wording changes did not trigger a new coding round. The three coders in Table 3 produced 97/113/106 spans; exact typed matches numbered 69/75/65 for A–B/A–C/B–C. Pairwise F1 ranges are 0.603–0.757 for K, 0.475–0.659 for S, and 0.722–0.837 for T. All coders identified the same two L spans in one sentence. O accounts for 59.5–64.5% of character positions. The linked reproduction materials retain the training records, freeze receipts, later-edit comparisons, per-type counts, and complete bootstrap specification; none of the 10,000 resamples produced an undefined reported coefficient.

APPENDIX C. SUPPLEMENTARY EVALUATION

This appendix documents the shared evaluation protocol, uncertainty and error diagnostics, and exploratory expanded-pool results. Scores and differences are calculated before rounding and displayed to three decimals; scientific notation preserves the sign of near-zero interval endpoints. The supporting records provide per-run ledgers, detailed scorer behaviour, and historical protocol comparisons. Those histories use different output formats and demonstration settings and therefore do not support controlled estimates of the effects of changing prompts.

Shared-Guideline Evaluation against the Human Reference

The shared protocol evaluates six configurations on all 150 reference sentences, using V4.2.12-based general instructions and five schematic examples.

Configuration and output records. Table 14 distinguishes the original annotation teacher, inference baselines, and expansion channels, with their recorded identifiers and access configurations. Models return an ID and spans as literal text, zero-based occurrence, and L/K/S/T type. The span parser resolves spans to Unicode offsets for exact and relaxed scoring with the released span scorer. All supplied input texts match the frozen reference.

Qwen uses deterministic decoding; Kimi uses non-thinking mode at temperature 0.6. DeepSeek output caps are 2,048 tokens without thinking and 8,192 with thinking. Access routes appear in Table 14; full configurations accompany the reproduction notes.

Parser and matching conventions. The parser resolves literal spans to source offsets, sorts candidates by start offset and original output order, and keeps the first compatible candidate. Later duplicates, same-boundary type conflicts, and overlaps are rejected. Partial responses retain accepted spans; fully rejected or malformed responses become empty predictions. Identifier coverage is checked across all supplied sentences before scoring.

Table 14. Recorded model identifiers, access routes, and roles in annotation and evaluation.

Role	Recorded identifier	Access and configuration
Original teacher	gpt-6-astra	Official OpenAI Codex; adopted Silver labels.
GPT (API)	gpt-5.6-terra	Third-party API gateway.
Claude (API)	Claude Sonnet 4.5	Third-party API gateway.
DeepSeek (API)	deepseek-v4-pro	Official DeepSeek API; non-thinking/thinking, 2,048/8,192-token output caps.
Kimi (API)	kimi-k2.6	Official Moonshot API; non-thinking; temperature 0.6.
Qwen baseline / LoRA	Qwen2.5-14B-Instruct	Local weights; deterministic decoding; output cap 2,048. Original Hub commit not recorded; archived file hashes identify the supplied artifacts.
JobBERT-zh	Released initialization and B2 continuation	Local encoder and inherited CRF; initialization-selection history incomplete.
Expansion annotation	OpenAI Codex / API gateway	Adopted Codex labels; gateway pool used only for supplementary comparison.

GPT and Claude used a third-party API gateway and are reported under service-recorded model names. No matched vendor-API comparison was available; their upstream model versions were not independently verified.

Exact matching deduplicates identical span tuples within each sentence. Relaxed matching traverses reference spans in source order and selects the unused same-type prediction with highest IoU, taking the first prediction on a tie. The threshold is 0.5, and matching is greedy rather than globally optimized. Empty–empty records contribute no span counts. The scorer specification preserves schema validation, BIO edge cases, and executable checks. The sentence-level counts reproduce the published exact scores before resampling.

Conditional resampling. We jointly resampled the 150 reference sentences across configurations 10,000 times with seed 20260923. Typed exact micro-F1 was recomputed from sentence-level span counts. For the LoRA mean, the same resampled sentences were used for all three fixed training seeds before averaging F1. Table 6 reports percentile intervals for paired differences. These post hoc intervals condition on the development-used reference and recorded runs. They do not model advertisement-level clustering or adjust for multiple comparisons. Seed SD instead describes variation across the recorded training runs.

Observed mismatch examples. In the frozen no-adaptor Qwen output for 1801–s0004, the predicted S span [5, 24) is 参与公司无人机飞控系统技术项目开发策划 (*participate in planning a technical development project for the company's UAV flight-control system*), whereas an overlapping reference S span is [16, 24), 技术项目开发策划 (*technical project development planning*). For 1802–s0005, both layers select 方法 (*method*) at [6, 8), but the prediction is S and the reference is K. These frozen outputs illustrate boundary and type mismatches with the reference; they have not been separately adjudicated. After removing exact matches, the recorded pairing procedure identifies 27 same-boundary/different-type pairs, 89 same-type/overlapping-boundary pairs, and 65 other unmatched predictions, totalling 181 false positives. These counts classify span relationships rather than linguistic causes. The mismatch ledger records the pairing order and parser errors; a full adjudicated error taxonomy remains unavailable.

Offline rescoring reproduces the exact and relaxed results for all six inference configurations and all three Qwen LoRA runs. The reproduction notes distinguish the scoring artifacts already available from the resources required to rerun model inference.

Under their respective output budgets, DeepSeek in thinking mode has higher precision and lower recall than the recorded non-thinking configuration.

Per-seed Qwen results. The per-seed record retains precision, recall, exact/relaxed F1, and calibration/challenge scores underlying Tables 7 and 6. Boundary exact F1 in Table 7 was subsequently recomputed from the same frozen accepted spans, with types collapsed and duplicate coordinates removed within sentences. This adds a scoring view without new training or inference. The model-diagnostic records provide per-seed and per-type precision, recall, TP/FP/FN, and the full-precision summaries used in Table 8; Figure 5 retains individual-seed Qwen results.

Matched Qwen Frozen-Prediction Diagnostics

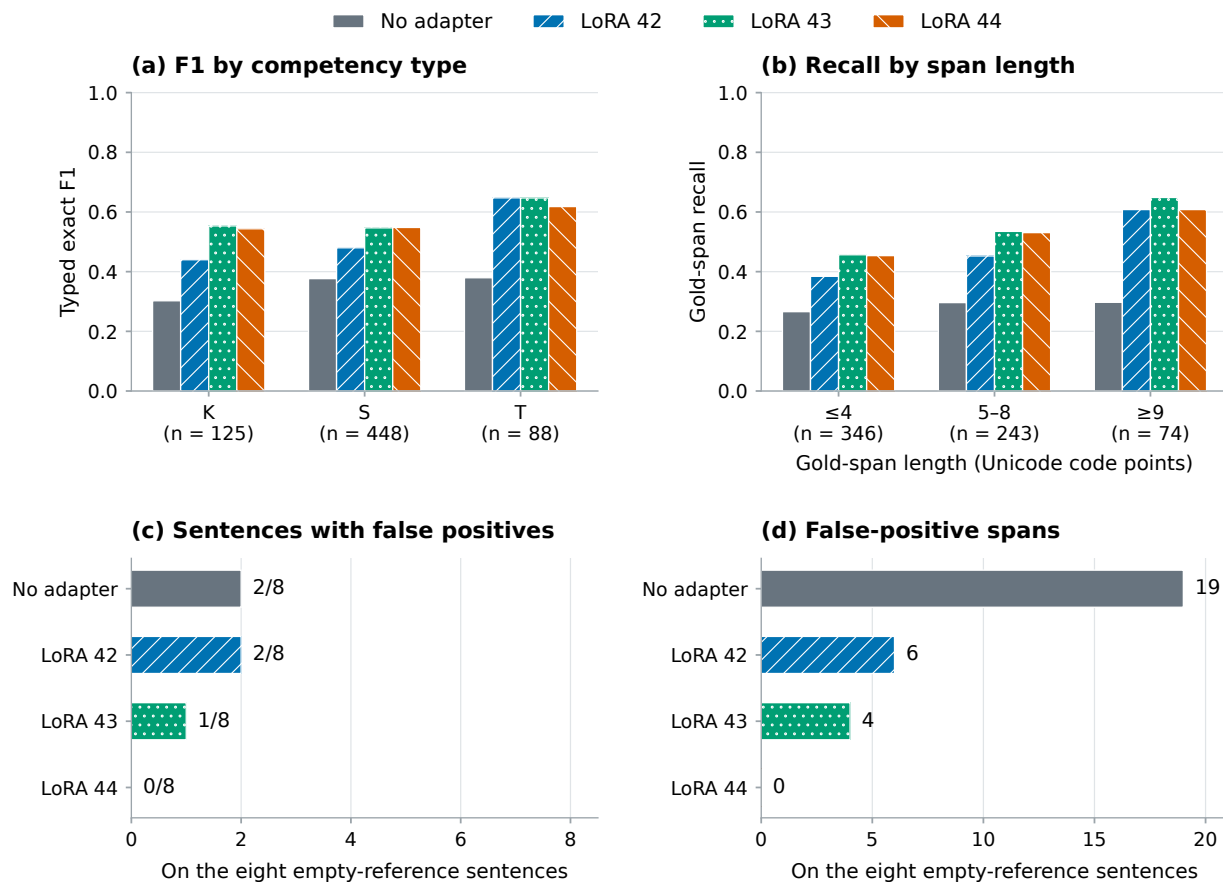


Figure 5. Qwen diagnostics on the same 150 reference sentences. (a) Typed exact F1 by competency type. (b) Recall by gold-span length; n counts gold spans. (c–d) False positives on eight empty-reference sentences, counted by sentence and span. Colour and pattern identify runs.

Diagnostics reuse the same 150 reference sentences and each run's frozen accepted spans. Typed exact matching is completed on each full sentence before aggregation by gold-span length. Bins (≤ 4 , 5–8, ≥ 9 Unicode code points) are fixed in advance; they contain 346/243/74 gold spans in

102/110/56 sentences, respectively. A sentence may contribute gold spans to multiple bins. Type-specific unmatched predictions count as FP even in empty-reference sentences. Results for each training seed are evaluated separately on the same reference sentences.

On the eight empty-reference sentences, accepted false-positive spans decrease from 19 without an adapter to 6, 4, and 0 for seeds 42, 43, and 44; affected sentences number 2, 2, 1, and 0.

L has two gold spans: the no-adapter run recovers 0/2 and each LoRA run 2/2, with one extra L prediction in every run. Complete TP/FP/FN, legal-empty-output counts, and parser-rejection diagnostics are retained in the diagnostic records.

Exploratory Expanded-Silver Experiments

Expanded-pool adaptation. The expanded Qwen runs use seed 42 and the shared prompt, occurrence output, parser, and Qwen training settings in Table 4. Loss applies only to assistant output. Both pooled checkpoints are selected at epoch one using Silver-validation exact F1; decoding is deterministic with a 2,048-token output cap. The complete configurations retain the parameter records. The expanded JobBERT runs retain the domain-adapted encoder and inherited CRF. Its latest run uses 8,842 training and 698 development records, so the separate 350-sentence Silver-test interpretation applies to Qwen, not to that JobBERT run.

Table 15 reports the two pooled Qwen runs against their held-out Silver targets and the shared human reference; the main results discuss the distinction between these evaluation targets.

Table 15. Expanded-pool Qwen results (seed 42). All scores are typed exact micro-F1 except the final relaxed-F1 column. Gold150 is used only for the additional human-reference evaluation.

Pool variant	Silver validation	Silver test	Gold150 exact	Gold150 relaxed
Alt.: 8,943 / 351 / 352	0.783	0.759	0.597	0.713
Codex: 8,842 / 348 / 350	0.833	0.809	0.585	0.703

Alt. is the supplementary alternative pool; Codex is adopted. Counts give train / validation / test sentences. Both retain earlier B2 and public-prompt annotations; names identify the channel for the replaced 2,500-sentence subset. Retained totals are 9,646 (Alt.) and 9,540 (Codex). Checkpoints were selected at epoch one with patience two. Differences use unrounded scores; rounding applies only to displayed values.

The Codex variant has higher agreement with its own Silver test labels but an approximately 0.011 lower exact F1 on Gold150 than the alternative variant. The channel replacement also changes record inclusion: 2,353 of the 2,500 Codex annotations are retained, compared with 2,459 alternative annotations. The contrast describes the resulting pools rather than an isolated annotation-channel effect.

The expanded JobBERT runs obtain Gold150 exact F1 of 0.242 for the alternative pool and 0.232 for the Codex pool (seed 42); the latter has relaxed F1 of 0.485. A preceding 7,117-record configuration gives 0.237 ± 0.002 across three seeds. These scores are lower than the historical B2 continuation result, but the expanded runs change the annotation protocol, source composition, and development set alongside training size. The latest JobBERT run selects on 698 development records, whereas Qwen uses 348 validation records and retains 350 for Silver testing. Thus these results do not identify an effect of increasing training size alone. They show that the larger supervision pool does not deliver uniform improvements across the recorded baselines; the matched Qwen comparison remains the direct evidence for adaptation under a fixed inference protocol. For comparison, the B2 Qwen adapter at seed 42 scores 0.500 on Gold150; its three-seed mean

of 0.540 is a different summary. Neither is a matched estimate of an expansion effect because the expanded runs change the supervision pool and selection targets. The lower expanded Job-BERT scores remain unexplained because the available records do not isolate label-version, input-windowing, or decoding effects. These runs remain exploratory. Exploratory supervision-subset runs and their selection arrangements are preserved in the supporting experiment records. These single-seed runs are not controlled estimates of training-size or annotation-channel effects.

APPENDIX D. REPRODUCTION AND VERSIONED MATERIALS

Table 16 summarizes the available materials and supported checks. The reproduction guide provides entry points for inspecting results, rescoring predictions, and rerunning models; its paper-to-file index maps tables and figures to supporting files.

The supporting record index links calculation details, per-run scores, and configurations to their protocols; the earlier process archive preserves exploratory subsets and version history. Qwen adapters and frozen predictions are available in the model-reproduction archive identified in Data Availability. These records support traceability but do not supply all training inputs or establish full reproduction. Table 16 distinguishes access and verification levels.

Collection and Preparation Records

Collection records name MacroData, operated by Chongqing Maherui Information Technology Co., Ltd., and the Recruitment Dataset (ID 163746) on Alibaba Cloud Tianchi.

The available public-institution records retain announcement titles, source names, URLs, listed publication dates, and geographic metadata where available. Listed publication dates are not collection dates. A complete source-by-stage census of advertisement counts, crawl windows, exclusions, and sentence-to-document mappings has not been recovered.

Audit of Frozen Expanded-Silver Splits

A post hoc audit of the complete frozen expanded-Silver files found that their hashes, counts, and identifier order match the archived split manifests. No identifiers or exact/NFC-normalized sentences are shared across partitions or with the 150-sentence human reference. An additional comparison using NFKC normalization, whitespace removal, and case folding identified the same single training-validation pair in both variants, differing only in full-width versus half-width semicolons; both records have empty span annotations. Three additional punctuation-variant pairs occur within each training set. No corresponding normalized-text overlap was found between training or validation and the Silver test partitions, or between any partition and the human reference. The frozen splits and reported results were retained; the audit does not rule out effects of validation overlap. Document-level and semantic near-duplicate isolation remain unestablished.

Resource Access and Sampling Scope

Expansion sampling. Preparation retained 2,064 valid unique sentences from the original supervision files and selected 7,936 new candidates: 3,174 cloud, 2,381 public-sector, and 2,381 listed-company sentences. Selection prioritized requirement-related keywords within each source and used fixed hash ranking with seed 20260913. Processing removed HTML and whitespace, split advertisements into sentence units, and kept normalized lengths of 8–400 characters. Matching sentences were excluded against existing corpus splits, supervision, human-review sets, and

Table 16. Component access and the reproduction level supported by the identified releases. Local possession is distinguished from public access.

Component	Access route / available objects	Scope and limitation
Core v0.1.3	Dataset archive: corpus, Gold150 (150), B2 (2,150/169), texts and labels	Download the named version; inspect labels and use its splits. It does not contain the later expanded pools.
Qwen reproduction archive	Model archive: three B2 adapters, selected expanded adapters, frozen predictions, prompts and scorer	Follow its model index for inference or rescoring; obtain base weights separately. Full training inputs are not bundled.
Expanded pools	9,540 adopted / 9,646 supplementary records; complete files verified locally against split manifests	IDs and selected outputs are archived; public complete texts and labels, or an executable controlled-access arrangement, remain unavailable. Not publicly retrainable as a complete study.
QA150 and historical coding	Pseudonymized texts and human layers linked in Data Availability; source hashes and cohort membership	Recalculate agreement without account metadata. The release preserves design distinctions and does not establish final-rule blind reliability.
Three-annotator blinded study	Agreement release: 50 sentences, frozen labels, source IDs, scorer and intervals	Recalculate independent agreement; five + 15 training sentences excluded. This is not a model test set.
Chinese encoder base-lines	Six public Hugging Face checkpoints; audit DOI 10.5281/zenodo.22937245	Frozen predictions, code and configurations support rescoring. Local rescoring verified all six runs; server checkpoint re-inference is separately recorded.
External experiments	Appendix E links public data/model sources and pinned result archives for 104 extraction and 36 additional classification/retrieval training runs	Metrics and available predictions support checks; unresolved provenance and environment gaps prevent a complete training-package claim.

Reuse terms vary by component. The authors' annotation permission excludes third-party text; separate model/base-model notices apply. Access on request and exact reconstruction from live recruitment pages are not guaranteed.

the available 3.2-million-sentence dump used for the paper's domain pretraining. The comparison key applies NFC, removes whitespace, and trims specified edge punctuation. Annotations use the prepared strings. These checks concern sentence matches under that key and do not establish advertisement-level or semantic near-duplicate isolation.

The preparation and audit records document the comparison-key function and checked pretraining dumps. The paper-main dump contains 3.2 million lines, but its content hash was not retained.

Two other local dumps, not used for exclusion, contain 143 and 4,919 selected-sentence matches.

The records do not resolve whether these other matches entered the reported pretraining run.

Annotation Channels and Frozen Targets

Both expanded pools retain earlier supervision and public-prompt components. The 2,500-candidate channel comparison and record exclusions yielded a supplementary gateway pool

Table 17. Recovered sampling-stage counts and missing census information. These are sentence counts, not advertisement counts or mutually exclusive industry totals.

Stage	Available counts	Sampling frame / unavailable detail
Original four collections	22,840; split 17,460/2,143/3,237	AI, Cloud, public-sector, graduate recruitment. Per-source ad counts, time windows, attrition and complete document mapping not recovered.
Targeted study lineage	980 conflict-origin + 1,621 other records	150 reference + 2,451 Silver-plus. This is a selected lineage, not a probability sample.
New expansion candidates	Cloud 3,174; public-sector 2,381; listed-company 2,381	Requirement-keyword-positive sentences; fixed hash ranking. Earlier B2 contributes 2,064 retained unique sentences.
Replacement channel subset	2,500 candidates; alternative retains 2,459; Codex retains 2,353	Respective exclusions: 41 (1.6%) and 147 (5.9%). A reason-coded attrition breakdown is not established.
Additional QA150	60 / 45 / 45 source quotas	Sampled from new Silver; machine suggestions visible. Does not extend the independent test population.

of 9,646 records and the adopted Codex pool of 9,540. The former remains named proxy in archived files. Adoption does not imply uniform model or handbook versions. The version and annotation records retain the stage-specific prompts, selection history, and original labels. A documented experience-boundary check concerns a single Silver example; it does not establish corpus-wide compatibility. Revised rules require a separately versioned annotation release.

Historical implementation. For some JobBERT runs, the exact tokenizer revision, special-token handling, truncation or windowing counts, and affected reference-span counts have not been recovered. The maximum input of 256 tokens is not a character limit, and input-length distributions alone cannot quantify truncation. The reported reference scores retain the full target sets. Per-seed checkpoint-selection records are stored with the available implementation history, including the inherited checkpoint's unresolved selection history.

APPENDIX E. EXTERNAL BENCHMARK DETAILS

Original-task Reruns and Scoring

We reran BERT (Devlin et al., 2019), SpanBERT (Joshi et al., 2020), JobBERT, and JobSpanBERT (Zhang et al., 2022b) separately for skills and knowledge on SkillSpan's public HOUSE and TECH subsets. The MaChAmp (van der Goot et al., 2021) seq_bio CRF runs used 20 epochs, development-set checkpoint selection, and five seeds listed in the experiment records. BIG was unavailable; multi-task training was not rerun. Table 19 gives MaChAmp per-site span-F1. The archived pooled scores report BIO rescoring of the same predictions with the correct task column. These scoring procedures yield different results. The native runs contain 3,570 test sentences, compared with 3,569 in the common-protocol release; the one-record difference has not been resolved by matching IDs. For published test references, we use only the original paper's TEST/STL rows. Its HOUSE/TECH rows report development results. For the German public ICT task (Gnehm et al., 2022b), jobBERT-de used a 20-epoch MaChAmp

configuration and ESCOXML-R used its released five-epoch configuration (Zhang et al., 2023). Both used five recorded seeds. Compatibility fields were added for newer MaChAmp without changing the learning rate or epoch limit; the exact changes are archived. The runs therefore use the public releases with compatibility changes, rather than recreating every original environment. They contain 2,557 test sentences, whereas the ESCOXML-R paper reports 2,943. ICT differs from the EDU/EXP/LNG extraction and taxonomy classification in Gnehm et al. (2022a).

Data and Implementation Sources

We obtained the common-protocol data from the public Hugging Face releases of SkillSpan, Kompetencer, Gnehm ICT, Green, Sayfullina, and FIJO. The corresponding papers are cited in the evaluation protocol; preprocessing and splits in these releases may differ from the original studies. The common protocol uses released BIO extraction streams: Sayfullina is a derived extraction task rather than its original candidate-phrase disambiguation task, and Kompetencer extraction is separate from the fine-grained classification of supplied spans reported below. Code is available from the SkillSpan, Kompetencer, ESCOXML-R, and MaChAmp repositories. The NNOSE repository supplies the retrieval-augmented implementation used in the subsequent A100 experiments. The preceding sections describe our linear-head protocol and compatibility changes. The experiment records below list the recorded revisions.

Uncertainty and Experiment Records

Table 18 gives the paired sentence-bootstrap intervals. Sayfullina's interval includes zero, although its lower endpoint is close to zero. Because the intervals are pointwise, exclusions of zero do not constitute significance tests adjusted for multiple comparisons. The intervals capture sentence sampling uncertainty conditional on the trained seeds; seed SD describes variation across runs. Neither accounts for dependence between sentences from the same advertisement.

The versioned experiment records link the common-protocol, Chinese adaptation, public-subset extraction, Kompetencer classification, and NNOSE retrieval experiments. They retain all seeds, model revisions, selected epochs, per-seed scores, and the available configuration and prediction hashes. Some dataset and executed-code identifiers remain unresolved.

External-task checks reconciled per-seed summaries with the ledgers; they did not rerun training or independently rescore predictions. For the Chinese encoders, local `cnss-lsht-1.2.0` rescoring reproduced all six runs' exact, relaxed, boundary-only, and per-type scores. Development-selected epochs and source offsets were also checked. Server-reported checkpoint re-inference matched archived tags on all 150 sentences per run; this was not a training rerun. The pinned audit archive and verification record preserve code, configurations, predictions, and checks.

Table 18. Paired differences under the common protocol (ESCOXLM-R minus XLM-R).

Dataset	Stream	Mean difference	95% interval
SkillSpan	Skill	+0.010	[−0.003, 0.023]
SkillSpan	Knowledge	−0.027	[−0.041, −0.013]
Kompetencer	Skill	−0.018	[−0.063, 0.022]
Kompetencer	Knowledge	−0.025	[−0.102, 0.056]
Gnehm ICT	ICT	−0.023	[−0.034, −0.013]
Green	Native types	−0.001	[−0.021, 0.017]
Sayfullina	Skill	+0.006	[−1.47 × 10 ^{−5} , 0.011]
FIJO	Native types	−0.004	[−0.043, 0.038]

Pointwise sentence-bootstrap intervals (2,000 resamples; seed 20260921), conditional on seeds 42/43/44. No advertisement clustering or multiple-comparison adjustment. Values are rounded after computing differences and intervals. Three decimal places are used except for Sayfullina’s near-zero lower endpoint, shown with three significant digits in scientific notation to preserve its negative sign. Its interval includes zero.

Table 19. SkillSpan test results: published single-task references and MaChAmp reruns on the public HOUSE and TECH subsets (0–1; five seeds).

Model	Task	Published TEST	Rerun HOUSE	Rerun TECH
BERT	Skills	0.543 ± 0.007	0.496 ± 0.013	0.514 ± 0.012
BERT	Knowledge	0.624 ± 0.004	0.550 ± 0.004	0.700 ± 0.008
SpanBERT	Skills	—	0.516 ± 0.014	0.526 ± 0.014
SpanBERT	Knowledge	—	0.536 ± 0.017	0.691 ± 0.015
JobBERT	Skills	0.561 ± 0.005	0.534 ± 0.014	0.538 ± 0.013
JobBERT	Knowledge	0.639 ± 0.003	0.577 ± 0.004	0.708 ± 0.005
JobSpanBERT	Skills	0.566 ± 0.008	0.543 ± 0.013	0.519 ± 0.010
JobSpanBERT	Knowledge	0.611 ± 0.010	0.546 ± 0.018	0.687 ± 0.014

Published values are the TEST/STL results in Table 5 of Zhang et al. (2022b), converted from percent and rounded to three decimals; their HOUSE/TECH rows report development results. Reruns use MaChAmp per-site span-F1 on public HOUSE+TECH, excluding BIG. The evaluation samples differ, so score differences cannot be interpreted as matched replication gaps. The source table gives no SpanBERT TEST score. Multi-task training was not rerun.

Table 20. German public ICT extraction: native-task recipes (0–1; five seeds).

Task/model	Recipe	Published	Rerun
Public ICT: jobBERT-de	20-epoch MaChAmp	—	0.885 ± 0.010
Public ICT: ESCOXLM-R	5-epoch released recipe	0.884 ± 0.005	0.891 ± 0.005

Reruns use 2,557 test sentences. The published ESCOXLM-R score comes from Table 2 of Zhang et al. (2023), whose Table 1 reports 2,943 test sentences. These different releases prevent a matched comparison with the published score. No corresponding published jobBERT-de score is supplied. Unlike the common-protocol comparison, these two native-task configurations use different training budgets measured in epochs.

Kompetencer Classification and NNOSE Reruns

Kompetencer fine-grained classification (Zhang et al., 2022a) and NNOSE (Zhang et al., 2024) were run on an NVIDIA A100 (80 GB). Software versions and compatibility records are provided

in the native-task configurations. These experiments retain task-specific metrics and are reported separately from the experiments using the common linear-head protocol. For Kompetencer, we compared DaJobBERT with RemBERT (Chung et al., 2021). MaChAmp used a 20-epoch budget and the five seeds recorded for the SkillSpan native runs. Training used Danish (DA), English (EN), or both (EN+DA), with 138 Danish and 9,472 English training records. All conditions selected checkpoints on the 1,577-record English development set. Table 21 reports unweighted macro-F1 for classification of supplied spans. The recorded DaJobBERT initialization is `jjzha/dajobbert-base-uncased`; the linked receipt retains its full revision. Together with the unweighted metric, this makes the experiment a task-specific rerun rather than a matched reproduction of the original cased setting. NNOSE used JobBERT and JobBERTa with seed 113412 and development-based early stopping. Each datastore contained only training examples from the evaluated dataset. Neighbour count k , interpolation weight λ , and temperature T were selected on development data and frozen before testing. Table 22 compares sequeval span-F1 from the linear head and retrieval-augmented predictions of the same checkpoint. These linear-head results belong to the NNOSE pipeline, not the common-protocol XLM-R comparison. The all-dataset datastore was not run, and the single-seed results do not establish a general retrieval benefit.

Table 21. Kompetencer fine-grained classification: macro-F1 across five seeds (0–1).

Encoder	Training	Danish test	English test
DaJobBERT	DA	0.182 ± 0.034	0.064 ± 0.012
DaJobBERT	EN	0.232 ± 0.028	0.539 ± 0.010
DaJobBERT	EN+DA	0.340 ± 0.017	0.529 ± 0.008
RemBERT	DA	0.050 ± 0.072	0.034 ± 0.041
RemBERT	EN	0.261 ± 0.019	0.552 ± 0.017
RemBERT	EN+DA	0.316 ± 0.019	0.554 ± 0.014

Mean $\pm$ sample SD; Danish/English test sizes are 784/1,578 classified spans. All conditions use English development data. EN training with Danish testing is the EN-to-DA transfer condition. DaJobBERT resolved to an uncased checkpoint despite the requested cased identifier. Scores are unweighted classification macro-F1, not span-F1 or the source plot’s weighted metric. No matched published-score comparison is claimed.

Table 22. NNOSE with an in-dataset training datastore: test span-F1 (0–1).

Dataset	Encoder	Test n	Linear	kNN
SkillSpan	JobBERT	3,569	0.613	0.612
SkillSpan	JobBERTa	3,569	0.631	0.625
Green	JobBERT	335	0.491	0.500
Green	JobBERTa	335	0.466	0.485
Sayfullina	JobBERT	1,851	0.897	0.896
Sayfullina	JobBERTa	1,851	0.915	0.912

Single seed 113412; no SD is estimated. Selected epochs and development-selected retrieval settings are preserved in the complete run table. Both columns use the same trained checkpoint within each row. The datastore excludes development and test examples. SkillSpan uses public HOUSE+TECH. The all-dataset datastore was not evaluated, and alignment with an original-paper result cell has not been established.

Combined English/Danish training improved Danish macro-F1 for both encoders; RemBERT’s

Danish-only score varied across seeds (0.050 ± 0.072). English-based selection and the un-weighted metric prevent a matched published-score comparison. NNOSE gave no consistent benefit: these single-seed runs improved Green but slightly reduced SkillSpan and Sayfullina F1. The coverage record lists uncompleted configurations; missing results are not scored as zero.

FUNDING

This research was supported by the National Social Science Fund of China under the project “Research on the Interaction Mechanism between Enterprise Employees’ Social Capital and Human Capital in the Intelligent Era and Its Impact on Career Success” (Grant No. 21BGL142). The funder had no role in designing the study, collecting and analysing the data, preparing the manuscript, or deciding whether to submit the article for publication.

COMPETING INTERESTS

The authors declare that they have no competing interests.

ACKNOWLEDGMENTS

The authors acknowledge the use of OpenAI Codex for manuscript revision and language editing, LaTeX formatting, and code preparation and checking. OpenAI’s image-generation tool assisted with the artwork in Figure 1. These tools were used under author direction; final decisions on manuscript content, analyses, interpretations, and conclusions remained with the authors. The authors take full responsibility for the manuscript, references, code, and figures, including verification of AI-assisted content. Models used in annotation and experiments are documented separately in the methods and reproduction materials. Preparation tools were used in September–October 2026; their model versions were not retained.

REFERENCES

- Artstein, R. and Poesio, M. (2008). Inter-coder agreement for computational linguistics. *Computational Linguistics*, 34(4):555–596.
- Beauchemin, D., Laumonier, J., Le Ster, Y., and Yassine, M. (2022). FIJO: a French Insurance Soft Skill Detection Dataset. In *Proceedings of the Canadian Conference on Artificial Intelligence*. Canadian Artificial Intelligence Association. Article 2022L34.
- Bender, E. M. and Friedman, B. (2018). Data statements for natural language processing: Toward mitigating system bias and enabling better science. *Transactions of the Association for Computational Linguistics*, 6:587–604.
- Bhola, A., Halder, K., Prasad, A., and Kan, M.-Y. (2020). Retrieving skills from job descriptions: A language model based extreme multi-label classification framework. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 5832–5842. International Committee on Computational Linguistics.
- Cao, L., Zhang, J., Ge, X., and Chen, J. (2021). Occupational profiling driven by online job advertisements: Taking the data analysis and processing engineering technicians as an example. *PLOS ONE*, 16(6):e0253308.

- Chen, C. and Kong, F. (2021). Enhancing entity boundary detection for better chinese named entity recognition. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, pages 20–25. Association for Computational Linguistics.
- Chen, X., Qin, C., Fang, C., Wang, C., Zhu, C., Zhuang, F., Zhu, H., and Xiong, H. (2024). Job-SDF: A multi-granularity dataset for job skill demand forecasting and benchmarking. In *Advances in Neural Information Processing Systems*, volume 37.
- Chung, H. W., Févry, T., Tsai, H., Johnson, M., and Ruder, S. (2021). Rethinking embedding coupling in pre-trained language models. In *International Conference on Learning Representations*.
- Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., and Stoyanov, V. (2020). Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451. Association for Computational Linguistics.
- Council of the European Union (2017). Council recommendation of 22 may 2017 on the european qualifications framework for lifelong learning. Official Journal of the European Union, C 189, pp. 15–28. Accessed: 2026-08-28.
- De Santo, A., Malandri, L., Mercorio, F., Mezzanzanica, M., and Nobani, N. (2026). SkillLens: Recognising and mapping novel skills from millions of job ads across europe using language models. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 5: Industry Track)*, pages 877–885. Association for Computational Linguistics.
- Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186. Association for Computational Linguistics.
- Ding, B., Qin, C., Liu, L., Chia, Y. K., Li, B., Joty, S., and Bing, L. (2023). Is GPT-3 a good data annotator? In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11173–11195. Association for Computational Linguistics.
- European Commission (2024). ESCO skills pillar: Skills and competences. https://esco.ec.europa.eu/en/classification/skill_main. Version 1.2.0 used to define the annotation inventory; accessed: 2026-08-28.
- Fetahu, B., Kar, S., Chen, Z., Rokhlenko, O., and Malmasi, S. (2023). SemEval-2023 task 2: Fine-grained multilingual named entity recognition (MultiCoNER 2). In *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023)*, pages 2247–2265. Association for Computational Linguistics.
- Geburu, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé III, H., and Crawford, K. (2021). Datasheets for datasets. *Communications of the ACM*, 64(12):86–92.
- Gnehm, A.-S., Bühlmann, E., Buchs, H., and Clematide, S. (2022a). Fine-grained extraction and classification of skill requirements in German-Speaking job ads. In *Proceedings of the Fifth Workshop on Natural Language Processing and Computational Social Science (NLP+CSS)*, pages 14–24. Association for Computational Linguistics.
- Gnehm, A.-S., Bühlmann, E., and Clematide, S. (2022b). Evaluation of transfer learning and

- domain adaptation for analyzing German-Speaking job advertisements. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 3892–3901, Marseille, France. European Language Resources Association.
- Green, T., Maynard, D., and Lin, C. (2022). Development of a benchmark corpus to support entity recognition in job descriptions. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 1201–1208, Marseille, France. European Language Resources Association.
- Gururangan, S., Marasović, A., Swayamdipta, S., Lo, K., Beltagy, I., Downey, D., and Smith, N. A. (2020). Don’t stop pretraining: Adapt language models to domains and tasks. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8342–8360. Association for Computational Linguistics.
- Hripcsak, G. and Rothschild, A. S. (2005). Agreement, the F-measure, and reliability in information retrieval. *Journal of the American Medical Informatics Association*, 12(3):296–298.
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. (2022). LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.
- Joshi, M., Chen, D., Liu, Y., Weld, D. S., Zettlemoyer, L., and Levy, O. (2020). SpanBERT: Improving pre-training by representing and predicting spans. *Transactions of the Association for Computational Linguistics*, 8:64–77.
- Kim, K. W., Kim, J.-D., and Aizawa, A. (2026). Refining and reusing annotation guidelines for LLM annotation. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 37951–37964. Association for Computational Linguistics.
- Klie, J.-C., Eckart de Castilho, R., and Gurevych, I. (2024). Analyzing dataset annotation quality management in the wild. *Computational Linguistics*, 50(3):817–866.
- Lafferty, J., McCallum, A., and Pereira, F. C. N. (2001). Conditional random fields: Probabilistic models for segmenting and labeling sequence data. In *Proceedings of the Eighteenth International Conference on Machine Learning*, pages 282–289. Morgan Kaufmann.
- le Vrang, M., Papantoniou, A., Pauwels, E., Fannes, P., Vandensteen, D., and De Smedt, J. (2014). ESCO: Boosting job matching in europe with semantic interoperability. *Computer*, 47(10):57–64.
- Li, M., Shi, T., Ziems, C., Kan, M.-Y., Chen, N., Liu, Z., and Yang, D. (2023). Coannotating: Uncertainty-guided work allocation between human and large language models for data annotation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 1487–1505. Association for Computational Linguistics.
- Lu, Y., Liu, Q., Dai, D., Xiao, X., Lin, H., Han, X., Sun, L., and Wu, H. (2022). Unified structure generation for universal information extraction. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5755–5772. Association for Computational Linguistics.
- Luo, H., Jin, Y., Wang, Y., Li, X., Shang, T., Liu, X., Chen, R., Wang, K., Salam, H., Wen, Q., and Liu, Z. (2025). DynamicNER: A dynamic, multilingual, and fine-grained dataset for LLM-based named entity recognition. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 16511–16535. Association for Computational Linguistics.
- MacroData (n.d.). MacroData. <https://www.macrodatas.cn/>. Platform operated by

- Chongqing Maherui Information Technology Co., Ltd. Accessed 20 September 2026.
- Magron, A., Dai, A., Zhang, M., Montariol, S., and Bosselut, A. (2024). JobScape: A framework for generating synthetic job postings to enhance skill matching. In *Proceedings of the First Workshop on Natural Language Processing for Human Resources (NLP4HR 2024)*, pages 43–58. Association for Computational Linguistics.
- Nguyen, K., Zhang, M., Montariol, S., and Bosselut, A. (2024). Rethinking skill extraction in the job market domain using large language models. In *Proceedings of the First Workshop on Natural Language Processing for Human Resources (NLP4HR 2024)*, pages 27–42. Association for Computational Linguistics.
- Peng, N. and Dredze, M. (2015). Named entity recognition for chinese social media with jointly trained embeddings. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 548–554.
- Podjanin, A. and Strietska-Ilina, O. (2020). The feasibility of using big data in anticipating and matching skills needs. Technical report, International Labour Organization, Geneva. Accessed: 2025-10-28.
- Qwen Team (2025). Qwen2.5 technical report. arXiv preprint arXiv:2412.15115v2. Version 2. Available at <https://arxiv.org/abs/2412.15115v2>.
- Sainz, O., García-Ferrero, I., Agerri, R., Lopez de Lacalle, O., Rigau, G., and Agirre, E. (2024). GoLLIE: Annotation guidelines improve zero-shot information-extraction. In *The Twelfth International Conference on Learning Representations*, volume 2024, pages 47083–47107.
- Sayfullina, L., Malmi, E., and Kannala, J. (2018). Learning representations for soft skill matching. In *Analysis of Images, Social Networks and Texts*, volume 11179 of *Lecture Notes in Computer Science*, pages 141–152. Springer.
- Senger, E., Zhang, M., van der Goot, R., and Plank, B. (2024). Deep learning-based computational job market analysis: A survey on skill extraction and classification from job postings. In *Proceedings of the Workshop on Natural Language Processing for Human Resources (NLP4HR)*, pages 1–15, St. Julian’s, Malta. Association for Computational Linguistics.
- Tianchi (n.d.). Recruitment Dataset. <https://tianchi.aliyun.com/dataset/163746/>. Title translated from Chinese; accessed 20 September 2026.
- Ul Haq, M. U., Rigoni, D., and Sperduti, A. (2026). A systematic comparison of large language models for data annotation in NER tasks. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference*, pages 524–548. ELRA Language Resource Association.
- van der Goot, R., Üstün, A., Ramponi, A., Sharaf, I., and Plank, B. (2021). Massive choice, ample tasks (MaChAmp): A toolkit for multi-task learning in NLP. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 176–197. Association for Computational Linguistics.
- Wang, S., Sun, X., Li, X., Ouyang, R., Wu, F., Zhang, T., Li, J., Wang, G., and Guo, C. (2025). GPT-NER: Named entity recognition via large language models. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 4257–4275. Association for Computational Linguistics.
- Xu, L., Yu tong, Dong, Q., Liao, Y., Yu, C., Tian, Y., Liu, W., Li, L., Liu, C., and Zhang, X. (2020). CLUENER2020: Fine-grained named entity recognition dataset and benchmark for Chinese.
- Zaratiana, U., Tomeh, N., Holat, P., and Charnois, T. (2024). GLiNER: Generalist model for named entity recognition using bidirectional transformer. In *Proceedings of the 2024 Confer-*

- ence of the North American Chapter of the Association for Computational Linguistics: *Human Language Technologies (Volume 1: Long Papers)*, pages 5364–5376, Mexico City, Mexico. Association for Computational Linguistics.
- Zhang, M., Jensen, K. N., and Plank, B. (2022a). Kompetencer: Fine-grained skill classification in danish job postings via distant supervision and transfer learning. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 436–447, Marseille, France. European Language Resources Association.
- Zhang, M., Jensen, K. N., Sonniks, S. D., and Plank, B. (2022b). SkillSpan: Hard and soft skill extraction from english job postings. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4962–4984, Seattle, United States. Association for Computational Linguistics.
- Zhang, M., Jensen, K. N., van der Goot, R., and Plank, B. (2022c). Skill extraction from job postings using weak supervision. In *Proceedings of the 2nd Workshop on Recommender Systems for Human Resources*, volume 3218 of *CEUR Workshop Proceedings*.
- Zhang, M., van der Goot, R., Kan, M.-Y., and Plank, B. (2024). NNOSE: Nearest neighbor occupational skill extraction. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 589–608. Association for Computational Linguistics.
- Zhang, M., van der Goot, R., and Plank, B. (2023). ESCOXML-R: Multilingual taxonomy-driven pre-training for the job market domain. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11871–11890. Association for Computational Linguistics.
- Zhou, W., Zhang, S., Gu, Y., Chen, M., and Poon, H. (2024). UniversalNER: Targeted distillation from large language models for open named entity recognition. In *The Twelfth International Conference on Learning Representations*.